

Jointly Mitigating Physical and Trotter Errors in Hamiltonian Simulation with Commutator Bounds

(experiment) + theory folks
(Dated:)

Product-formula Hamiltonian simulation is attractive for near-term quantum devices because of its simple circuit structure and commutator-sensitive error scaling, but its performance is limited by both algorithmic Trotter error and physical noise. Existing mitigation strategies typically handle these two errors separately, requiring extrapolation over both the Trotter step size and the noise strength. We develop a one-dimensional joint mitigation protocol that couples the tunable noise strength to the Trotter step size as $\lambda(s) = c(sT)^{p+1}$ for a p -th order product formula, so that the leading physical-noise and Trotter contributions appear at the same order and can be canceled by a single Richardson extrapolation. The analysis uses a finite-order BCH truncation bound controlled by doubly right-nested commutators, avoiding the need for convergence of the full BCH series and yielding commutator-sensitive guarantees for noisy product formulas. We instantiate the bounds for k -local Hamiltonians and second-quantized plane-wave electronic-structure Hamiltonians, and show that under sufficiently mild local Lindbladian noise the physical noise can be mitigated together with the Trotter error with only a constant asymptotic overhead compared with noiseless Trotter extrapolation. Numerical simulations further demonstrate that the proposed one-dimensional procedure can achieve accuracy comparable to a two-dimensional Richardson baseline while using fewer extrapolation points.

Summary of the results:

1. Provide a general condition involving the doubly right-nested commutators of Hamiltonians, where the Trotter error mitigation algorithm [1] achieves commutator and logarithm precision scaling, and show that k -local Hamiltonians and the second-quantized plane-wave electronic structure Hamiltonians satisfy the condition.
2. Show that under mild physical noise, the Trotter error mitigation algorithm can simultaneously mitigate the physical noise incurring a constant complexity overhead. **Yicong:** In addition to the number of interpolation nodes, try to prove that the overall sample complexity of ZNE procedure via 1D extrapolation is smaller than that by 2D extrapolation. **Shuo:** 2D complexity to be added, even though maybe mild improvement.

All of the algorithms use at most two ancilla qubits. **Shuo:** Confused: I doubt if this is the cost of lindbladian simulation, Trotter needs no ancilla.

I. INTRODUCTION

Hamiltonian simulation is one of the most promising candidates to demonstrate practical quantum advantage on near-term devices [2], with applications ranging from condensed matter physics [3], quantum chemistry [4], to high-energy physics [5].

The first explicit quantum simulation algorithm was proposed by Lloyd [6] using product formulas [7, 8], and then numerous advanced techniques have been extensively developed, including linear combination of unitaries (LCU) [9, 10], quantum signal processing

(QSP) [11, 12], etc. Despite the inferior cost dependence in time t and precision ϵ , the product formula enjoys commutator scaling and does not require ancillary qubits or complex controlled gates, enabling its convenient implementation on near-term quantum devices.

Watson and Watkins [1] exponentially reduced the error dependence of Trotter using extrapolation. However, providing fully rigorous commutator bounds for general Hamiltonians is challenging. A key technical difficulty is that the convergence of the BCH formula, which the analysis relies on, is not guaranteed in all regimes (e.g., for general k -local Hamiltonians), making the derivation of commutator bounds difficult. Important progress was recently made by Mizuta [13], who provided a rigorous commutator bound for the specific case of lattice Hamiltonians by using a subsystem Trotterization technique. Furthermore, our recent work [14] derives an upper bound of the truncation error of the BCH formula for general Lindbladians in terms of the doubly right-nested commutators, which naturally provides a rigorous error bound for Trotter extrapolation for general Hamiltonians, including lattice and second-quantized Hamiltonians even with dissipation.

Besides the Trotter error, physical noise is another central obstruction on near-term devices. Quantum error correction provides a scalable route to fault tolerance [15], but it requires resources far beyond what is currently available in near-term experiments. Quantum error mitigation (QEM) [16], including zero-noise extrapolation (ZNE) [17, 18], probabilistic error cancellation [19, 20], and measurement error mitigation [21, 22], is more immediately compatible with noisy devices, though it can increase the number of circuit samples needed for a fixed accuracy.

Most existing approaches treat algorithmic Trotter error and physical noise separately, for example by combin-

ing ZNE with an extrapolation in the Trotter step size or conversely, performing a two-dimensional extrapolation [23]. Recent work showed that the two errors can instead be coupled and mitigated by a one-dimensional extrapolation [24]. However, the existing analysis focuses on first-order Trotter formulas, uses an approximate global depolarizing-noise model, and does not provide commutator-sensitive bounds. This motivates the central question addressed in this work:

Can we jointly mitigate physical and Trotter errors while maintaining commutator scaling?

Tongyang: to be extended The main contribution of this work is to answer this question by applying the Lindbladian BCH truncation analysis of Ref. [14] to noisy product formulas. By coupling the noise strength λ to the Trotter step size s as $\lambda = c(sT)^{p+1}$ for a p -th order product formula, the leading physical-noise and Trotter contributions occur at the same order in s and can be removed by a single Richardson extrapolation. We prove commutator-sensitive bounds for this one-dimensional joint mitigation scheme, including applications to k -local lattice Hamiltonians and second-quantized plane-wave electronic-structure Hamiltonians. Under sufficiently mild intrinsic noise, the physical noise can be mitigated together with the Trotter error with only a constant asymptotic overhead.

II. PRELIMINARIES

In this section, we introduce the notations, key concepts, and backbone lemmas used throughout the paper.

A. Notations

We denote the set of all positive integers by $\mathbb{Z}_+$. For a sequence of operators $\mathcal{A}_1, \dots, \mathcal{A}_\Gamma$, we denote $\prod_{\gamma=1}^\Gamma \mathcal{A}_\gamma = \mathcal{A}_\Gamma \mathcal{A}_{\Gamma-1} \dots \mathcal{A}_1$ and $\prod_{\gamma=\Gamma}^1 \mathcal{A}_\gamma = \mathcal{A}_1 \mathcal{A}_2 \dots \mathcal{A}_\Gamma$. We denote the right-nested commutator $[X_1, [X_2, [\dots [X_{n-1}, X_n] \dots]]$ by $[X_1, X_2, \dots, X_n]$. We denote $\{1, 2, \dots, n\}$ by $[n]$ and the collection of all size- m subsets of $[n]$ by $\binom{[n]}{m}$. We use $\text{supp}(X)$ to denote the set of qubits that X nontrivially acts on. For a superoperator $\mathcal{L}$, we denote the diamond norm of $\mathcal{L}$ by $\|\mathcal{L}\|$. Define the time-ordering operator $\mathcal{T}\{\cdot\}$ as

$$\mathcal{T}\{[O_1(t_1), \dots, O_q(t_q)]\} = [O_{\pi(1)}(t_{\pi(1)}), \dots, O_{\pi(q)}(t_{\pi(q)})],$$

where $\pi: [q] \rightarrow [q]$ is a permutation of $[q]$ such that

$$t_{\pi(1)} \geq t_{\pi(2)} \geq \dots \geq t_{\pi(q)}.$$

For any permutation $\sigma: [q] \rightarrow [q]$, define the reordering operator $\mathcal{R}_\sigma\{\cdot\}$ as

$$\mathcal{R}_\sigma\{[O_1(t_1), \dots, O_q(t_q)]\} = [O_{\sigma(1)}(t_{\sigma(1)}), \dots, O_{\sigma(q)}(t_{\sigma(q)})].$$

Throughout this paper, we denote the Lindblad dissipator with a jump operator L as

$$\mathcal{D}[L](\rho) := L\rho L^\dagger - \frac{1}{2}\{L^\dagger L, \rho\}.$$

B. Quantum error mitigation

Quantum error mitigation (QEM) [16] refers to algorithmic schemes that reduce the noise-induced bias in the observable expectation value by post-processing outputs from an ensemble of circuit runs. QEM has enabled the quantum hardware in the noisy intermediate-scale quantum era to achieve practical applications [25].

1. Zero-noise extrapolation

One of the most common QEM techniques is *zero-noise extrapolation* (ZNE) [17, 18], which tries to obtain the zero-noise value using data points at different circuit fault rates. Specifically, we use ρ_λ to denote the output quantum state at a given circuit fault rate λ , and model the observable average $\text{Tr}[O\rho_\lambda]$ using an ansatz function $f(\lambda, \vec{\theta})$ with a set of parameters $\vec{\theta}$. Starting from the smallest circuit fault rate λ_1 that we can achieve, we boost the fault rate to obtain a set of data points $\{\lambda_m, \text{Tr}[O\rho_{\lambda_m}]\}$. We then fit the ansatz to the data points to choose a set of optimal parameters $\vec{\theta}^*$. The zero-noise observable average $\text{Tr}[O\rho_0]$ is estimated by

$$\text{Tr}[O\rho_0] \approx f(0, \vec{\theta}^*).$$

If λ is *small*, Ref. [18] showed that $\text{Tr}[O\rho_\lambda]$ can be approximated using a polynomial ansatz in the same spirit as truncated Taylor series:

$$\text{Tr}[O\rho_\lambda] \approx f(\lambda; \vec{\theta}) = \sum_{\ell=0}^{M-1} \theta_\ell \frac{\lambda^\ell}{\ell!},$$

where the $(M-1)$ -degree polynomial is determined by M different free parameters and $f(0, \vec{\theta}^*) = \theta_0^*$. In general, the number of data points λ_m is greater than or equal to M . For the simplest case where $M = 2$ [17], we can perform linear regression. When the number of data points reaches its minimum needed amount M , we can perform Richardson extrapolation as discussed in [Appendix A 1](#). **Shuo: In practice, when using product formulas, we can only consider odd or even polynomials as ansatzes.**

A straightforward approach to boost the fault rate λ is *unitary folding*, which replaces a quantum circuit or gate U by $U(U^\dagger U)^n$ [26–28]. This increases the noise rate by a factor of $1 + 2n$ without knowing the noise model. However, the circuit depth will also increase accordingly, exceeding the favorable regime for near-term devices. Alternatively, if we can completely characterize the noise model a priori, we can probabilistically insert gates to

simulate the faults directly [17]. On the other hand, knowing a concrete noise model enables us to adopt more precise ansatzes. For instance, one can rigorously prove that amplifying global depolarizing noise via unitary folding leads to the scaling form $\text{Tr}[O\rho_\lambda] = a + bp^\lambda$ [26].

C. Truncation error of the BCH formula

Let $\mathcal{K}_1, \dots, \mathcal{K}_M$ be general Lindbladians. The BCH formula can then be written as

$$e^{\mathcal{K}_M} \dots e^{\mathcal{K}_2} e^{\mathcal{K}_1} = \exp\left(\sum_{q=1}^{\infty} \Phi_q\right), \quad (1)$$

where each Φ_q is a sum of the weighted right-nested commutators of $\mathcal{K}_j$ (see Arnal et al. [29]):

$$\Phi_q = \sum_{\substack{p_v \geq 0, \\ p_1 + \dots + p_M = q}} \frac{1}{p_1! \dots p_M!} \cdot \sum_{\sigma \in S_q} c_{\sigma,q} \mathcal{R}_\sigma \left\{ \underbrace{[\mathcal{K}_M, \dots, \mathcal{K}_M]_{p_M}}_{p_M}, \dots, \underbrace{[\mathcal{K}_1, \dots, \mathcal{K}_1]_{p_1}}_{p_1} \right\}, \quad (2)$$

where S_q denotes the set of permutations of $[q]$,

$$c_{\sigma,q} = \frac{1}{q^2} \frac{(-1)^{d_\sigma}}{\binom{q-1}{d_\sigma}}, \quad (4)$$

satisfying $|c_{\sigma,q}| \leq 1/q^2$, and d_σ denotes the number of $i \in [q-1]$ such that $\sigma(i) > \sigma(i+1)$.

Define the sum of the q -fold right-nested commutator norms as

$$\alpha_{\text{comm}}^{(q)} = \sum_{v_1, \dots, v_q=1}^M \|[\mathcal{K}_{v_1}, \dots, \mathcal{K}_{v_q}]\|. \quad (5)$$

Aftab et al. [30] showed that the norm of Φ_q can be bounded as

$$\|\Phi_q\| \leq \frac{1}{q^2} \alpha_{\text{comm}}^{(q)},$$

hence, the series expansion in Eq. (2) converges if there exist two constants $J, C \geq 0$ such that

$$\sup_{q \geq J} \alpha_{\text{comm}}^{(q)} \leq C. \quad (6)$$

A trivial bound for $\alpha_{\text{comm}}^{(q)}$ is

$$\alpha_{\text{comm}}^{(q)} \leq \left(2 \sum_{v=1}^M \|\mathcal{K}_v\|\right)^q, \quad (7)$$

by $\|[A, B]\| \leq 2\|A\|\|B\|$ and the triangle inequality. The quantity $\alpha_{\text{comm}}^{(q+1)}$ also appears in the complexity of q -th

order Trotterization [31], where each $\mathcal{K}_v$ is a Hermitian operator in the decomposition of the simulated Hamiltonian. For some physical Hamiltonians, such as k -local Hamiltonians on a lattice of size N , where each $\mathcal{K}_v$ can be decomposed as a sum of terms acting nontrivially on at most k sites, $\alpha_{\text{comm}}^{(q)}$ admits a sharper bound than Eq. (7). This leads to the commutator scaling of the q -th order Trotterization. Specifically, suppose that the maximum interaction strength of $\sum_{v=1}^M \mathcal{K}_v$ on each site is g . Eq. (7) yields $(2Ng)^q$, while the bound considering the commutator structure of $\mathcal{K}_v$ gives $Nq!(2kg)^q = O(N(qkg)^q)$. The latter bound gives better complexity scaling for local Hamiltonians where $k \ll N$ and q is a constant.

However, in the error analysis of some advanced Trotter methods with higher precision [1, 30], bounds on $\alpha_{\text{comm}}^{(q)}$ for a single q do not suffice. Instead, we require the BCH formula to be convergent. The convergence condition in Eq. (6) requires $\alpha_{\text{comm}}^{(q)}$ to be bounded for arbitrarily large q , and is thus too strict to utilize the commutator structure of $\mathcal{K}_v$, since the only valid choice to satisfy the condition is $g = 0$. **XZ: add connection to trotter, and comparision to previous analysis** Concretely, in Trotter error mitigation [1], one expands the extrapolation function $f(s)$ as a power series around $s = 0$, whose validity relies on the convergence of the underlying BCH series. For a k -local system, each generator satisfies $\mathcal{K}_v = O(s)$, yielding $\alpha_{\text{comm}}^{(q)} = O(s^q q! (kg)^q)$. The factorial growth dominates the s^q suppression, so the BCH series has zero radius of convergence: for any $s > 0$, the series diverges. Consequently, the power expansion of $f(s)$ is merely a formal asymptotic series rather than a convergent one. This contradicts the implicit analyticity assumption in the asymptotic analysis and is precisely the gap that makes the prior analyses [1, 30] lack full rigor for general k -local systems. Our work [14] resolves this issue via the truncated BCH formula (**Theorem 1**), which avoids the convergence requirement by working directly with a finite-order truncation and bounding the remainder, yielding rigorous error bounds even outside the convergence radius. The situation is the same if each $\mathcal{K}_v$ is a k -local Lindbladian, as the parameter $\alpha_{\text{comm}}^{(q)}$ admits similar bounds (see **Corollary 1**).

To bypass the divergence issue of the BCH formula, Mizuta [13] showed that, for k -local Hamiltonians on a lattice, a truncated BCH formula can approximate the exponential product even outside the convergence radius of the full BCH series. Their argument avoids the convergence requirement by working directly with a finite truncation, but it relies on a subsystem Trotterization technique that is specific to lattice Hamiltonians. For the general setting considered here, we instead use the truncation theorem of Ref. [14], which is proved by constructing an equivalent continuous-time evolution and applying the Magnus expansion. This approach reduces the truncation problem to bounding the following doubly right-nested commutators.

Define the sum of doubly right-nested commutator norms as

$$\bar{\alpha}_{\text{comm}}^{(q_1, \dots, q_d)} = \sum_{v_{1,1}, \dots, v_{d,q_d}=1}^M \left\| [\mathcal{K}_{v_{1,1}}, \dots, \mathcal{K}_{v_{1,q_1}}], \dots, [\mathcal{K}_{v_{d,1}}, \dots, \mathcal{K}_{v_{d,q_d}}] \right\|. \quad (8)$$

These quantities are tailored to the finite-order BCH truncation error. For k -local systems, the doubly nested structure can be bounded by products of local factors $(2kg)^{q_r}$ and support-growth factors P_{r+1} , rather than by a global norm bound such as Eq. (7). For second-quantized plane-wave electronic-structure Hamiltonians, an analogous layer-counting argument yields bounds of order $\prod_r P_{r+1} q_r! n^{q_r}$. These factorized estimates are strong enough to make the truncated BCH error small in regimes where the full BCH series is not guaranteed to converge.

We first state the general truncation theorem, and then estimate its commutator parameter for the concrete Hamiltonian classes of interest. In the following theorem, we recall the BCH truncation bound that allows us to control the product-formula error without assuming convergence of the full BCH series.

Theorem 1 ([14, Theorem 6], BCH Truncation Bound). *For any positive integer q_0 , define $\bar{\alpha}_{\text{comm}, q_0}$ as*

$$\bar{\alpha}_{\text{comm}}^{(q_0+1)} + \sum_{r=0}^{\infty} \frac{1}{(r+1)!} \sum_{\substack{1 \leq p_1, \dots, p_{r+1} \leq q_0 \\ p_1 + \dots + p_{r+1} \geq q_0 + 2}} \bar{\alpha}_{\text{comm}}^{(p_1, \dots, p_{r+1})}. \quad (9)$$

If $\bar{\alpha}_{\text{comm}, q_0} \leq 1$, the truncation error of the q_0 -th order BCH formula is bounded as

$$\left\| \exp \left(\sum_{q=1}^{q_0} \Phi_q \right) - \prod_{v=1}^M e^{\mathcal{K}_v} \right\| \leq e \bar{\alpha}_{\text{comm}, q_0}.$$

Shuo: We demonstrate the power of Theorem 1 by applying it to the k -local and the second-quantized electronic Hamiltonians in Appendix B.

III. ANALYSIS OF THE EXTRAPOLATION ERROR

Roadmap. In this section, we derive the 1D joint error mitigation scheme and analyze the extrapolation error. Specifically, we upper bound the extrapolation error by the linear combination of the simulation errors of each noisy quantum circuit, which can be further decomposed into the BCH truncation error of the noisy quantum channel plus the distance between the truncated BCH formula and the ideal evolution channel.

Given a Hamiltonian $H = \sum_{\gamma=1}^{\Gamma} H_{\gamma}$, an initial state ρ_0 , and an observable O , we aim to estimate

$$\text{Tr}[O e^{-i \text{ad}_H T} \rho_0],$$

for any simulation time T . Suppose that the Hamiltonian evolution driven by each H_{γ} can be efficiently implemented by a layer of gates. Then, for sufficiently small t , we can implement the Hamiltonian evolution e^{-iHt} approximately by a p -th order product formula of the form

$$P(t) = \prod_{v=1}^M e^{-i a_v H_{\gamma_v}},$$

where the product is ordered from right to left. Here, a_v are real coefficients, $M = \Upsilon \Gamma$ is the total number of unitary gates, and the sequence γ_v specifies which Hamiltonian is applied at step v . Denote $a_{\max} = \max_{v \in [M]} |a_v|$. The corresponding ideal channel $\mathcal{P}(t)$ is a superoperator given by

$$\mathcal{P}(t) = \prod_{v=1}^M e^{-i a_v \text{ad}_{H_{\gamma_v}}},$$

where $\mathcal{A}_{\gamma} := \text{ad}_{H_{\gamma}}$ denotes the adjoint action of H_{γ} .

Suppose that the system is subject to physical noise with a controllable strength λ . Specifically, after each layer of unitary gates, a noise channel $e^{\lambda \mathcal{L}_v}$ is applied, where $\mathcal{L}_v$ is a Lindbladian. The resulting noisy channel is

$$\tilde{\mathcal{P}}(t, \lambda) = \prod_{v=1}^M (e^{\lambda \mathcal{L}_v} e^{-i a_v \text{ad}_{H_{\gamma_v}}}).$$

In experimental implementations, the noise strength is lower-bounded by the device's intrinsic noise level, denoted by $\lambda_{\min}$. To access larger noise strengths, one common approach is *unitary folding* [26], which replaces each unitary gate U by $U(U^{\dagger}U)^n$, effectively scaling the noise by a factor of $1 + 2n$ without requiring explicit knowledge of the noise model. However, folding increases the circuit depth proportionally, which may exceed the coherence budget of near-term devices. Alternatively, if the noise channel can be characterized—for instance, learned as a sparse Pauli-Lindblad model [20]—one can probabilistically insert recovery gates to directly emulate amplified noise at the pulse level without additional circuit depth [17]. In practice, the noise strength cannot be amplified arbitrarily without degrading the signal, but moderate amplification is experimentally feasible. We therefore assume that λ can be tuned within the range $[\lambda_{\min}, \infty)$ for the purpose of asymptotic analysis. Given $s \in (0, 1)$ such that $1/s \in \mathbb{Z}$, we implement $\mathcal{P}(sT)^{1/s}$ with initial state ρ_0 and then measure the observable O to approximate $\text{Tr}[O e^{-i \text{ad}_H T} \rho_0]$. We denote the expectation of the measurement results with noisy circuit by

$$g(s, \lambda) := \text{Tr}[O \tilde{\mathcal{P}}(sT, \lambda)^{1/s} \rho_0],$$

and note that

$$\lim_{s, \lambda \rightarrow 0} g(s, \lambda) = \text{Tr}[Oe^{-i\text{ad}_H T} \rho_0]$$

is the quantity we want. To obtain this limit value, one could perform two successive extrapolations: first, use ZNE to extrapolate $\lambda \rightarrow 0$ for a fixed s , and then use Trotter error mitigation [1] to extrapolate $s \rightarrow 0$ using the ZNE output. Although this method is possible, it involves extrapolating two independent parameters, which requires more sample points. Furthermore, each extrapolation step amplifies other sources of noise, such as state preparation and measurement (SPAM) errors, making the method less robust.

Therefore, we consider a more efficient and robust approach: a unified 1D extrapolation. By coupling the noise strength and the step size, we reduce this 2D problem to a single-variable extrapolation. In general, we could set λ to be an analytic function of s , $\lambda(s)$, such that $\lambda(0) = 0$. This ensures that the extrapolated function $g(s, \lambda(s))$ has the correct limit value at $s = 0$. Furthermore, its analyticity near $s = 0$ guarantees the existence of a power series expansion, which is the fundamental requirement for Richardson extrapolation. For general analytic function $\lambda(s) = \sum_{j=p_0+1}^{\infty} c_j (sT)^j$, the noise error will be dominated by its leading-order term, $O(s^{p_0+1})$. Thus, for analyzing the asymptotic scaling, it is sufficient to consider a simple monomial form $\lambda(s) = c(sT)^{p_0+1}$ for some $c > 0$ and $p_0 \in \mathbb{Z}_+$.

We now determine the optimal exponent p_0 . The total error of the estimator $g(s, \lambda(s))$ arises from two dominant sources: the algorithmic (Trotter) error, which scales as $O(s^{p+1})$ for a p -th order product formula, and the physical noise error, which scales linearly with λ in the leading order, and thus scales as $O(s^{p_0+1})$. The leading-order error of the combined system is governed by the term with the slower decay, i.e., $O(s^{\min\{p+1, p_0+1\}})$. To maximize the efficiency of the extrapolation, we aim to maximize this leading-order exponent. If we choose $p_0 < p$, the noise error dominates ($O(s^{p_0+1})$) and becomes the bottleneck, wasting the precision of the p -th order Trotter formula. Conversely, if $p_0 > p$, the Trotter error dominates ($O(s^{p+1})$), implying that we are suppressing the noise strength more aggressively than necessary. The optimal strategy is to match the scaling of the two error sources by setting $p_0 = p$. This choice ensures that both the leading-order Trotter error and the leading-order noise error appear at the same order, $O(s^{p+1})$, allowing the 1D Richardson extrapolation to eliminate both simultaneously and efficiently. We further remark that this matching strategy is not merely a heuristic. As we will demonstrate in our applications (see Section III A and Section III B), this choice enables our protocol to achieve a simulation depth that saturates the fundamental limits of quantum error mitigation under Pauli noise, as established in Refs. [32, 33].

To analyze the Richardson extrapolation error, our goal is to bound the remainder of the power series ex-

pansion of

$$f(s) := g(s, c(sT)^{p+1}) = \text{Tr}[O\tilde{\mathcal{P}}(sT, c(sT)^{p+1})^{1/s} \rho_0].$$

The $1/s$ exponent in the definition of $f(s)$ makes a direct power series analysis difficult. The standard approach is to first analyze the single-step propagator, $\tilde{\mathcal{P}}(sT, c(sT)^{p+1})$, by finding its effective generator. Note that the full channel can be written as

$$\tilde{\mathcal{P}}(t, \lambda) = \prod_{v=1}^{2M} e^{\mathcal{K}_v},$$

where each $\mathcal{K}_v$ is a Lindbladian defined as

$$\mathcal{K}_j = \begin{cases} -ita_v \text{ad}_{H_{\gamma_v}} & \text{if } j = 2v - 1 \\ \lambda \mathcal{L}_v & \text{if } j = 2v \end{cases}, \quad (10)$$

for $v \in [M]$. Therefore, the effective generator of $\tilde{\mathcal{P}}(t, \lambda)$ is given by the BCH formula

$$\tilde{\mathcal{P}}(t, \lambda) = \exp\left(\sum_{q=1}^{\infty} \Phi_q(t, \lambda)\right),$$

where Φ_q is the q -th order term in the BCH formula in Eq. (2) with $2M$ generators $\mathcal{K}_1, \dots, \mathcal{K}_{2M}$ defined in Eq. (10). The full evolution is then

$$\tilde{\mathcal{P}}(sT, c(sT)^{p+1})^{1/s} = \exp\left(\frac{1}{sT} \sum_{q=1}^{\infty} \Phi_q(sT, c(sT)^{p+1})T\right).$$

The analysis of the power series expansion of $\tilde{\mathcal{P}}(sT, c(sT)^{p+1})^{1/s}$ reduces to analyzing the scaled effective generator

$$\frac{1}{sT} \sum_{q=1}^{\infty} \Phi_q(sT, c(sT)^{p+1}).$$

To resolve the divergence issue of this BCH series as discussed in Section II C, we consider the truncated scaled effective generator

$$\mathcal{G}_{(q_0)}(s) := \frac{1}{sT} \sum_{q=1}^{q_0} \Phi_q(sT, c(sT)^{p+1}), \quad (11)$$

where the truncation order q_0 is determined by Theorem 1 so that error induced by the truncation, $\|\exp(\mathcal{G}_{(q_0)}(s)T) - \tilde{\mathcal{P}}(sT, c(sT)^{p+1})^{1/s}\|$, is within the required error tolerance. The Richardson extrapolation error is therefore determined by the power series of the truncated scaled effective generator $\mathcal{G}_{(q_0)}(s)$.

To analyze the power series of $\mathcal{G}_{(q_0)}(s)$, we first show that each $\Phi_q(t, \lambda)$ is a homogeneous polynomial of t and λ and bound the coefficients of the polynomial. For ease of presentation, we also define the generators without coefficients as

$$\tilde{\mathcal{K}}_j := \begin{cases} \text{ad}_{H_{\gamma_v}} & \text{if } j = 2v - 1 \\ \mathcal{L}_v & \text{if } j = 2v \end{cases}, \quad (12)$$

for $v \in [M]$.

Lemma 1 (BCH Coefficient Bound). *Define the sum of the commutator norms for the base generators $\tilde{\mathcal{K}}_j$ as*

$$\alpha_{\text{comm}}^{(q_1, q_2)} = \sum_{(j_1, \dots, j_q) \in S_{q_1, q_2}} \|\tilde{\mathcal{K}}_{j_1}, \tilde{\mathcal{K}}_{j_2}, \dots, \tilde{\mathcal{K}}_{j_q}\|, \quad (13)$$

$$S_{q_1, q_2} := \left\{ (j_1, \dots, j_q) \in [2M]^q \mid |\{k \in [q] \mid j_k \text{ is odd}\}| = q_1, |\{k \in [q] \mid j_k \text{ is even}\}| = q_2 \right\}.$$

The q -th order term $\Phi_q(t, \lambda)$ in the BCH formula of $\tilde{\mathcal{P}}(t, \lambda)$ can be written as a homogeneous polynomial of t and λ :

$$\Phi_q(t, \lambda) = \sum_{\substack{q_1, q_2 \geq 0 \\ q_1 + q_2 = q}} \Phi_{q_1, q_2} t^{q_1} \lambda^{q_2}, \quad (14)$$

such that the coefficients Φ_{q_1, q_2} are bounded by

$$\|\Phi_{q_1, q_2}\| \leq \frac{a_{\text{max}}^{q_1}}{q^2} \alpha_{\text{comm}}^{(q_1, q_2)}. \quad (15)$$

Proof. Substituting Eq. (10) into the expression of the q -th order BCH term $\Phi_q(t, \lambda)$ (Eq. (2)) yields

$$\begin{aligned} \Phi_q(t, \lambda) = & \sum_{\substack{p_v, p'_v \geq 0 \\ p_1 + p'_1 + \dots + p_M + p'_M = q}} \left(\prod_{v=1}^M \frac{\lambda^{p_v} (-i a_v)^{p'_v}}{p_v! p'_v!} \right) \\ & \cdot \sum_{\sigma \in S_q} c_{\sigma, q} R_{\sigma} \left\{ \underbrace{\mathcal{L}_M, \dots, \mathcal{L}_M}_{p_M}, \underbrace{\mathcal{A}_{\gamma_M}, \dots, \mathcal{A}_{\gamma_M}}_{p'_M}, \dots, \underbrace{\mathcal{L}_1, \dots, \mathcal{L}_1}_{p_1}, \underbrace{\mathcal{A}_{\gamma_1}, \dots, \mathcal{A}_{\gamma_1}}_{p'_1} \right\}. \end{aligned} \quad (16)$$

Note that any summand in Eq. (16) containing $[\tilde{\mathcal{K}}_{j_1}, \tilde{\mathcal{K}}_{j_2}, \dots, \tilde{\mathcal{K}}_{j_q}]$ for $(j_1, \dots, j_q) \in [2M]^q$ is a monomial of λ and t . The degree of t is $|\{k \in [q] \mid j_k \text{ is odd}\}| =: q_1$ and the degree of λ is $|\{k \in [q] \mid j_k \text{ is even}\}| =: q_2$. We now bound the coefficient of this monomial. The commutator $[\tilde{\mathcal{K}}_{j_1}, \tilde{\mathcal{K}}_{j_2}, \dots, \tilde{\mathcal{K}}_{j_q}]$ can only originate from one summand in the sum over p_v, p'_v in Eq. (16) because the operator counts $(p_1, p'_1, \dots, p_M, p'_M)$ are uniquely determined by $(j_1, \dots, j_q)$. The number of permutations $\sigma \in S_q$ in the second sum in Eq. (16) that preserve the commutator is

$$\prod_{v=1}^M p_v! p'_v!.$$

Thus, the coefficient of the monomial $t^{q_1} \lambda^{q_2} [\tilde{\mathcal{K}}_{j_1}, \dots, \tilde{\mathcal{K}}_{j_q}]$ can be upper bounded by

$$\begin{aligned} & \left\| \frac{1}{q^2} \prod_{v=1}^M \frac{(-i a_v)^{p'_v}}{p_v! p'_v!} \prod_{v=1}^M p_v! p'_v! \|\tilde{\mathcal{K}}_{j_1}, \tilde{\mathcal{K}}_{j_2}, \dots, \tilde{\mathcal{K}}_{j_q}\| \right\| \\ & \leq \frac{a_{\text{max}}^{q_1}}{q^2} \|\tilde{\mathcal{K}}_{j_1}, \tilde{\mathcal{K}}_{j_2}, \dots, \tilde{\mathcal{K}}_{j_q}\|, \end{aligned}$$

as $|c_{\sigma, q}| \leq 1/q^2$. Then, we regroup the sum in Eq. (16) according to $(j_1, \dots, j_q) \in S_{q_1, q_2}$ (which have the same

degree of t and λ). The resulting polynomial

$$\Phi_q(t, \lambda) = \sum_{\substack{q_1, q_2 \geq 0 \\ q_1 + q_2 = q}} \Phi_{q_1, q_2} t^{q_1} \lambda^{q_2}$$

satisfies

$$\begin{aligned} \|\Phi_{q_1, q_2}\| & \leq \frac{a_{\text{max}}^{q_1}}{q^2} \sum_{(j_1, \dots, j_q) \in S_{q_1, q_2}} \|\tilde{\mathcal{K}}_{j_1}, \tilde{\mathcal{K}}_{j_2}, \dots, \tilde{\mathcal{K}}_{j_q}\| \\ & = \frac{a_{\text{max}}^{q_1}}{q^2} \alpha_{\text{comm}}^{(q_1, q_2)}. \end{aligned}$$

□

By Lemma 1, $\mathcal{G}_{(q_0)}(s)$ is a power series in s and we denote the q -th order coefficient by $\mathcal{E}_{(q_0), q+1}$ such that

$$\mathcal{G}_{(q_0)}(s) = \sum_{q=0}^{\infty} \mathcal{E}_{(q_0), q+1} (sT)^q. \quad (17)$$

We bound the norm of $\mathcal{E}_{(q_0), q}$ in the following lemma.

Lemma 2. *Define a constant*

$$\mu_{\text{comm}, q_0} = \max_{q' \geq p+1} \left(\sum_{\substack{1 \leq q_1 + q_2 \leq q_0 \\ q_1 + (p+1)q_2 = q'}} \frac{a_{\text{max}}^{q_1} c^{q_2}}{(q_1 + q_2)^2} \alpha_{\text{comm}}^{(q_1, q_2)} \right)^{1/q'}. \quad (18)$$

Then we have

$$\mathcal{G}_{(q_0)}(s) = -\text{iad}_H + \sum_{q=p}^{\infty} \mathcal{E}_{(q_0),q+1}(sT)^q, \quad (19)$$

where $\mathcal{E}_{(q_0),q}$ can be bounded by

$$\|\mathcal{E}_{(q_0),q}\| \leq \mu_{\text{comm},q_0}^q. \quad (20)$$

Proof. By the order condition of the product formula (see [1, Lemma 2]), we have

$$\frac{1}{sT} \sum_{q=1}^{\infty} \Phi_q(sT, 0) = -\text{iad}_H + O((sT)^p),$$

which implies

$$\Phi_{q_1,0} = 0,$$

for all $1 < q_1 \leq p$, and

$$\Phi_{1,0} = -\text{iad}_H sT.$$

Therefore, for any $c > 0$, we have

$$\begin{aligned} & \frac{1}{sT} \Phi_q(sT, c(sT)^{p+1}) \\ &= \frac{1}{sT} \left(\sum_{q_1=1}^{q_0} \Phi_{q_1,0}(sT)^{q_1} \right. \\ & \quad \left. + \sum_{q=1}^{q_0} \sum_{\substack{q_1 \geq 0, q_2 \geq 1 \\ q_1 + q_2 = q}} \Phi_{q_1,q_2}(sT)^{q_1} (c(sT)^{p+1})^{q_2} \right) \end{aligned}$$

$$\begin{aligned} &= -\text{iad}_H + \sum_{q_1=p+1}^{q_0} \Phi_{q_1,0}(sT)^{q_1-1} + O((sT)^p) \\ &= -\text{iad}_H + \sum_{q=p}^{\infty} \mathcal{E}_{(q_0),q+1}(sT)^q. \end{aligned}$$

For any $q \geq p+1$, we have

$$\begin{aligned} \|\mathcal{E}_{(q_0),q}\| &= \left\| \sum_{q'=1}^{q_0} \sum_{\substack{q_1, q_2 \geq 0, \\ q_1 + q_2 = q'}} \Omega_{q_1, q_2} c^{q_2} \right\| \\ &\leq \sum_{\substack{1 \leq q_1 + q_2 \leq q_0 \\ q_1 + (p+1)q_2 = q}} \frac{a_{\max}^{q_1} c^{q_2}}{(q_1 + q_2)^2} \alpha_{\text{comm}}^{(q_1, q_2)} \\ &\leq \mu_{\text{comm},q_0}^q. \end{aligned}$$

□

Lemma 3. The evolution $e^{\mathcal{G}_{(q_0)}(s)T}$ admits series expansion in s given by

$$e^{\mathcal{G}_{(q_0)}(s)T} = e^{-\text{iad}_H T} + \sum_{q=p}^{\infty} \tilde{\mathcal{E}}_{(q_0),q} s^q,$$

where $\tilde{\mathcal{E}}_{(q_0),q}$ are superoperators bounded by

$$\|\tilde{\mathcal{E}}_{(q_0),q}\| \leq e(2\mu_{\text{comm},q_0} T)^{q(1+1/p)},$$

if $2\mu_{\text{comm},q_0} T \geq 1$, and bounded by

$$\|\tilde{\mathcal{E}}_{(q_0),q}\| \leq e(2\mu_{\text{comm},q_0} T)^q,$$

if $2\mu_{\text{comm},q_0} T \leq 1$.

Proof. By the Dyson series expansion in the interaction picture, we have

$$\begin{aligned} & e^{\mathcal{G}_{(q_0)}(s)T} - e^{-\text{iad}_H T} \\ &= e^{-\text{iad}_H T} \sum_{j=1}^{\infty} \int_0^T d\tau_1 \int_0^{\tau_1} d\tau_2 \cdots \int_0^{\tau_{j-1}} d\tau_j \prod_{j'=j}^1 \left(e^{\text{iad}_H \tau_{j'}} \left(\sum_{q=p}^{\infty} \mathcal{E}_{(q_0),q+1} s^q T^q \right) e^{-\text{iad}_H \tau_{j'}} \right) \\ &= e^{-\text{iad}_H T} \sum_{q=p}^{\infty} s^q \sum_{j=1}^{\infty} \sum_{\substack{q_1, \dots, q_j \geq p \\ q_1 + \dots + q_j = q}} \int_0^T d\tau_1 \int_0^{\tau_1} d\tau_2 \cdots \int_0^{\tau_{j-1}} d\tau_j \prod_{j'=j}^1 \left(e^{\text{iad}_H \tau_{j'}} \mathcal{E}_{(q_0),q_{j'}+1} e^{-\text{iad}_H \tau_{j'}} T^{q_{j'}} \right) \\ &=: \sum_{q=p}^{\infty} \tilde{\mathcal{E}}_{(q_0),q} s^q. \end{aligned}$$

The superoperator $\tilde{\mathcal{E}}_{(q_0),q}$ can be bounded by

$$\begin{aligned}
\|\tilde{\mathcal{E}}_{(q_0),q}\| &\leq \sum_{j=1}^{\infty} \sum_{\substack{q_1, \dots, q_j \geq p \\ q_1 + \dots + q_j = q}} \int_0^T d\tau_1 \int_0^{\tau_1} d\tau_2 \cdots \int_0^{\tau_{j-1}} d\tau_j \prod_{j'=1}^j (\|\mathcal{E}_{(q_0),q_{j'}+1}\| T^{q_{j'}}) \\
&\leq \sum_{j=1}^{\infty} \sum_{\substack{q_1, \dots, q_j \geq p \\ q_1 + \dots + q_j = q}} \frac{T^{q+j}}{j!} \prod_{j'=1}^j \mu_{\text{comm},q_0}^{q_{j'}+1} \\
&= \sum_{j=1}^{\lfloor q/p \rfloor} \frac{(\mu_{\text{comm},q_0} T)^{q+j}}{j!} \left(\sum_{\substack{q_1, \dots, q_j \geq p \\ q_1 + \dots + q_j = q}} 1 \right).
\end{aligned}$$

The second sum can be bounded by

$$\sum_{\substack{q_1, \dots, q_j \geq p \\ q_1 + \dots + q_j = q}} 1 \leq \sum_{\substack{q_1, \dots, q_j \geq 0 \\ q_1 + \dots + q_j = q}} 1 = \binom{q+j-1}{j-1} \leq 2^{q+j}.$$

Therefore, for $2\mu_{\text{comm},q_0} T \geq 1$, we have

$$\begin{aligned}
\|\tilde{\mathcal{E}}_{(q_0),q}\| &\leq \sum_{j=1}^{\lfloor q/p \rfloor} \frac{(2\mu_{\text{comm},q_0} T)^{q+j}}{j!} \\
&\leq (2\mu_{\text{comm},q_0} T)^{q+q/p} \sum_{j=1}^{\infty} \frac{1}{j!} \\
&\leq e(2\mu_{\text{comm},q_0} T)^{q(1+1/p)},
\end{aligned}$$

and for $2\mu_{\text{comm},q_0} T \leq 1$, we have

$$\begin{aligned}
\|\tilde{\mathcal{E}}_{(q_0),q}\| &\leq (2\mu_{\text{comm},q_0} T)^q \sum_{j=0}^{\infty} \frac{(2\mu_{\text{comm},q_0} T)^j}{j!} \\
&\leq (2\mu_{\text{comm},q_0} T)^q e^{2\mu_{\text{comm},q_0} T} \\
&\leq e(2\mu_{\text{comm},q_0} T)^q.
\end{aligned}$$

□

We now combine the BCH truncation bound, the effective-generator expansion, and Richardson extrapolation to obtain a general resource bound for the joint mitigation protocol.

Theorem 2. Let $H = \sum_{\gamma=1}^{\Gamma} H_{\gamma}$ be the target Hamiltonian. Suppose we approximate its evolution by implementing a p -th order product formula with M gate layers:

$$P(t) = \prod_{v=1}^M e^{-it a_v H_{\gamma_v}},$$

and that the implementation is subject to physical noise described by $e^{\lambda \mathcal{L}_v}$ after the v -th gate layer, with a tunable noise strength $\lambda \geq \lambda_{\min}$.

Let $c > 0$ and $q_0 \in \mathbb{Z}_+$. Let $\bar{\alpha}_{\text{comm},q_0}(s)$ be the doubly right-nested commutator bound (Eq. (9)) evaluated for the generators:

$$\mathcal{K}_j = \begin{cases} -isT a_v \text{ad}_{H_{\gamma_v}} & \text{if } j = 2v - 1 \\ c(sT)^{p+1} \mathcal{L}_v & \text{if } j = 2v \end{cases}.$$

Let μ_{comm,q_0} be the parameter defined in Eq. (18).

There exists an algorithm that, given any observable O and initial state ρ_0 , estimates $\text{Tr}[Oe^{-\text{iad}_H T}\rho_0]$ to within an additive error of $\epsilon\|O\|$ using a total of

$$O\left(\max\left\{\frac{\log^3(1/\epsilon)}{\log^2(1/\epsilon)s_0^{-1}}, \frac{(\log \log(1/\epsilon))^3}{\epsilon^2}\right\}\right) = \tilde{O}\left(\frac{1}{s_0\epsilon^2}\right)$$

Trotter steps. This involves $\tilde{O}(1/\epsilon^2)$ independent circuit runs, each comprising at most

$$O(\max\{\log^3(1/\epsilon), \log^2(1/\epsilon)s_0^{-1}\}) = \tilde{O}\left(\frac{1}{s_0}\right)$$

Trotter steps, where s_0 is the largest step size satisfying

$$s_0(2\mu_{\text{comm},q_0}T)^{1+1/p} \leq 1/e, \quad 2e\bar{\alpha}_{\text{comm},q_0}(s_0) \leq s_0 \frac{\epsilon}{4eC \log m}, \quad (21)$$

with C being the Richardson constant and $m = O(\log(1/\epsilon))$ satisfying $2e^{1-m} \leq \epsilon/(4C \log m)$.

This result holds provided that the intrinsic noise strength of the device, $\lambda_{\min}$, satisfies the feasibility condition $\lambda_{\min} = \tilde{O}(c(s_0T)^{p+1})$.

Proof. Recall that the function for extrapolation is

$$f(s) = \text{Tr}[O(\tilde{\mathcal{P}}(sT, c(sT)^{p+1}))^{1/s}\rho_0],$$

for $s > 0$ and $1/s \in \mathbb{Z}$. Here we consider the scenario where $2\mu_{\text{comm},q_0}T \geq 1$. By [Lemma 3](#), the evolution $e^{\mathcal{G}_{(q_0)}(s)T}$ admits series expansion in s given by

$$e^{\mathcal{G}_{(q_0)}(s)T} = e^{-\text{iad}_H T} + \sum_{q=p}^{\infty} \tilde{\mathcal{E}}_{(q_0),q} s^q,$$

such that

$$\|\mathcal{E}_{(q_0),q}\| \leq e(2\mu_{\text{comm},q_0}T)^{q(1+1/p)}.$$

Then the function $f(s)$ can be decomposed as

$$\begin{aligned} f(s) &= \text{Tr}[Oe^{\mathcal{G}_{(q_0)}(s)T}\rho_0] + \text{Tr}[O((\tilde{\mathcal{P}}(sT, c(sT)^{p+1}))^{1/s} - e^{\mathcal{G}_{(q_0)}(s)T})\rho_0] \\ &= P_m(s) + R_m(s), \end{aligned}$$

$$P_m(s) := \text{Tr}[Oe^{-\text{iad}_H T}\rho_0] + \sum_{q=p}^{m-1} \text{Tr}[O\mathcal{E}_{(q_0),q}\rho_0]s^q,$$

$$R_m(s) := \sum_{q=m}^{\infty} \text{Tr}[O\mathcal{E}_{(q_0),q}\rho_0]s^q + \text{Tr}[O(\tilde{\mathcal{P}}(sT, c(sT)^{p+1}) - e^{\mathcal{G}_{(q_0)}(s)T})\rho_0].$$

where P_m is a degree- $(m-1)$ polynomial in s . Let $s_1, \dots, s_m$ and $b_1, \dots, b_m$ be the extrapolation points and coefficients defined in [Lemma 7](#) with $t = 1$ and $s = s_0$ to be determined later, and then we have $1 \leq \|\mathbf{b}\| \leq C \log m$, $1/s_j \in \mathbb{Z}$, and $1/s_j = r_j = O(\max\{m^3, m^2/s\}/j^2)$. We can set

$$m = O(\log(1/\epsilon)),$$

such that

$$2e^{1-m} \leq \frac{\epsilon}{4C \log m}. \quad (22)$$

For any s satisfying

$$\begin{aligned} e\bar{\alpha}_{\text{comm},q_0}(s) &\leq s \frac{\epsilon}{4eC \log m} \leq s, \\ s(2\mu_{\text{comm},q_0}T)^{1+1/p} &\leq 1/e, \end{aligned} \quad (23)$$

the remainder $R_m(s)$ is bounded as

$$\begin{aligned} |R_m(s)| &\leq \sum_{q=m}^{\infty} \|O\| \|\tilde{\mathcal{E}}_{(q_0),q}\| \|\rho_0\|_1 s^q + \|O\| \left\| (\tilde{\mathcal{P}}(sT, c(sT)^{p+1}))^{1/s} - (e^{\mathcal{G}_{(q_0)}(s)T})^{1/s} \right\| \|\rho_0\|_1 \\ &\leq \sum_{q=m}^{\infty} e(s(2\mu_{\text{comm},q_0}T)^{1+1/p})^q \|O\| + \frac{(\max\{1, \|e^{\mathcal{G}_{(q_0)}(s)T}\|\})^{1/s}}{s} \left\| \tilde{\mathcal{P}}(sT, c(sT)^{p+1}) - e^{\mathcal{G}_{(q_0)}(s)T} \right\| \|O\| \end{aligned}$$

$$\begin{aligned}
&\leq 2e^{1-m}\|O\| + \frac{(1 + e\alpha_{\text{comm},q_0}(s))^{1/s}}{s} e\alpha_{\text{comm},q_0}(s)\|O\| \\
&\leq \frac{\varepsilon}{4C\log m}\|O\| + (1+s)^{1/s} \frac{\varepsilon}{4eC\log m}\|O\| \\
&\leq \frac{\varepsilon}{2C\log m}\|O\|,
\end{aligned}$$

where the third line follows from [Lemma 3](#) and [Lemma 9](#), the fourth line follows from [Theorem 1](#), and the fifth line follows from Eq. (22) and Eq. (23).

Now, we bound the error of the extrapolation function

$$F^{(m)} = \sum_{j=1}^m b_j f(s_j).$$

Set

$$f(0) := \lim_{s \rightarrow 0} f(s) = \text{Tr}[Oe^{-i\text{ad}_H T} \rho_0].$$

Let s_0 be the largest number smaller than $e^{-1}(2\mu_{\text{comm},q_0}T)^{-1-1/p}$ and satisfying

$$e\bar{\alpha}_{\text{comm},q_0}(s_0) \leq s_0 \frac{\varepsilon}{4eC\log m}.$$

Observe that $\bar{\alpha}_{\text{comm},q_0}(s)$ consists of sums of norms of nested commutators involving $-isTa_v\mathcal{A}_{\gamma_v}$ and $c(sT)^{p+1}\mathcal{L}_v$ with grade $q \geq q_0 + 1$. Consequently, for $s \geq 0$, it is a polynomial in s with non-negative coefficients and a lowest degree of $q_0 + 1$. This implies that the function $\bar{\alpha}_{\text{comm},q_0}(s)/s$ is monotonically increasing for $s \geq 0$. By [Lemma 7](#), the extrapolation nodes satisfy $s_j \leq s_0$. Therefore, as the condition in Eq. (23) holds for the base step size s_0 , it is satisfied for all s_j . Then we have

$$\begin{aligned}
|F^{(m)}(s) - f(0)| &\leq \|\mathbf{b}\|_1 \max_j |R_m(s_j)| \\
&\leq C\log m \frac{\varepsilon}{2C\log m}\|O\| \\
&\leq \frac{\varepsilon}{2}\|O\|.
\end{aligned}$$

To estimate $f(s_j)$ with error $\varepsilon\|O\|/(2\|\mathbf{b}\|_1)$ with success probability $1 - 1/(3m)$, we need to repeatedly run $(\tilde{\mathcal{P}}(s_jT, c(s_jT)^{p+1}))^{1/s_j}$ $O(\|\mathbf{b}\|_1^2/\varepsilon^2 \log m)$ times. Each application of $(\tilde{\mathcal{P}}(s_jT, c(s_jT)^{p+1}))^{1/s_j}$ requires $1/s_j$ Trotter steps. Hence, the maximum number of Trotter steps per coherent run is

$$\begin{aligned}
O(\max_j r_j) &= O(\max\{m^3, m^2s_0^{-1}\}) \\
&= O(\max\{\log^3(1/\varepsilon), \log^2(1/\varepsilon)s_0^{-1}\}) \\
&= \tilde{O}\left(\frac{1}{s_0}\right),
\end{aligned}$$

and the total number of Trotter steps is

$$N_{\text{Trot}} = O\left(\sum_{j=1}^m r_j \frac{\|\mathbf{b}\|_1^2}{\varepsilon^2} \log m\right)$$

$$\begin{aligned}
&= O\left(\sum_j^m \frac{\max\{m^3, m^2s_0^{-1}\} \log^3 m}{j^2 \varepsilon^2}\right) \\
&= O\left(\max\{\log^3(1/\varepsilon), \log^2(1/\varepsilon)s_0^{-1}\} \frac{(\log \log(1/\varepsilon))^3}{\varepsilon^2}\right) \\
&= \tilde{O}\left(\frac{1}{s_0\varepsilon^2}\right).
\end{aligned}$$

The smallest noise strength is

$$\begin{aligned}
\min_j c(s_jT)^{p+1} &= \Omega\left(\min_j c\left(\frac{j^2T}{\max\{m^3, m^2s_0^{-1}\}}\right)^{p+1}\right) \\
&= \Omega\left(c \min\left\{\frac{T^{p+1}}{\log^{3p+3}(1/\varepsilon)}, \frac{T^{p+1}s_0^{p+1}}{\log^{2p+2}(1/\varepsilon)}\right\}\right) \\
&= \tilde{\Omega}(c(s_0T)^{p+1}).
\end{aligned}$$

□

XZ: TBD: explaining three tunable parameters: c balances Trotter error and physical error, q_0 balances BCH truncation error and extrapolation error. The algorithm involves three tunable parameters that control different aspects of the error budget:

- The coupling constant c in $\lambda(s) = c(sT)^{p+1}$ determines the relative weight of the physical noise error versus the Trotter error in the leading-order term. In [Section III A](#), we optimize c analytically (Eq. (37)) to balance the two sources so that neither dominates the extrapolation remainder.
- The BCH truncation order q_0 controls the truncation error through $\bar{\alpha}_{\text{comm},q_0}$ (see [Theorem 1](#)). A larger q_0 pushes the BCH remainder to higher order, while it can also increase the commutator parameter μ_{comm,q_0} in the effective-generator expansion. In the general theorem, q_0 is chosen so that the second condition in Eq. (21) holds. Its scaling depends on the commutator-tail estimate available for the Hamiltonian and noise class. For the k -local application below, it suffices to choose $q_0 = O(\log(N/\epsilon))$.
- The number of extrapolation nodes m (equivalently, the minimum step size s_0) governs the Richardson extrapolation accuracy. By [Lemma 7](#), the extrapolation error decays exponentially in m , so $m = O(\log(1/\epsilon))$ nodes are sufficient.

Shuo: Variance-sensitive sampling refinement. The shot-complexity estimate above uses a worst-case bounded-outcome concentration argument. A sharper statement keeps the single-shot variances of the extrapolation data points. We record this refinement separately because it clarifies how a one-dimensional path can have a smaller statistical prefactor than a two-dimensional grid, even though both retain the same leading $1/\varepsilon^2$ dependence in the worst case.

Lemma 4 (Bernstein bound with shot allocation). *Let*

$$\hat{F} = \sum_{i=1}^m a_i \hat{O}_i, \quad \hat{O}_i = \frac{1}{n_i} \sum_{\ell=1}^{n_i} X_{i,\ell},$$

where the random variables $X_{i,\ell}$ are independent and satisfy $\mathbb{E}X_{i,\ell} = O_i$, $\text{Var}(X_{i,\ell}) = \sigma_i^2$, and $|X_{i,\ell}| \leq \|O\|$. Then

$$\Pr\{|\hat{F} - \mathbb{E}\hat{F}| \geq \varepsilon\} \leq 2 \exp\left(-\frac{\varepsilon^2}{2V + \frac{4}{3}\|O\|\varepsilon B}\right), \quad (24)$$

where

$$V = \sum_{i=1}^m \frac{a_i^2 \sigma_i^2}{n_i}, \quad B = \max_i \frac{|a_i|}{n_i}.$$

If every data point receives the same number of shots, the total number of shots sufficient for statistical error at most ε with failure probability at most δ is

$$N_{\text{eq}} = O \left[\left(\frac{m \sum_{i=1}^m a_i^2 \sigma_i^2}{\varepsilon^2} + \frac{\|O\| m \max_i |a_i|}{\varepsilon} \right) \log \frac{1}{\delta} \right]. \quad (25)$$

Moreover, by allocating shots according to the Cauchy–Schwarz, or Neyman, rule for the variance term, with an additional floor proportional to $|a_i|$ to control the range term, one obtains

$$N_{\text{opt}} = O \left[\left(\frac{(\sum_{i=1}^m |a_i| \sigma_i)^2}{\varepsilon^2} + \frac{\|O\| \|\mathbf{a}\|_1}{\varepsilon} \right) \log \frac{1}{\delta} \right]. \quad (26)$$

Proof. Eq. (24) follows by applying Bernstein’s inequality to the centered variables $a_i(X_{i,\ell} - O_i)/n_i$. Their total variance is V , and each summand has absolute value at most $2\|O\||a_i|/n_i$.

For equal shots, $n_i = N_{\text{tot}}/m$, and therefore $V = m \sum_i a_i^2 \sigma_i^2 / N_{\text{tot}}$ and $B = m \max_i |a_i| / N_{\text{tot}}$, which gives Eq. (25). For the optimized allocation, let $S = \sum_i |a_i| \sigma_i$. Allocate one part of the shot budget with $n_i^{(v)} \propto |a_i| \sigma_i$ and one part with $n_i^{(r)} \propto |a_i|$. The first part gives $V \leq S^2 / N_v$ by Cauchy–Schwarz, while the second gives $B \leq \|\mathbf{a}\|_1 / N_r$. Choosing $N_v = O(S^2 \varepsilon^{-2} \log(1/\delta))$ and $N_r = O(\|O\| \|\mathbf{a}\|_1 \varepsilon^{-1} \log(1/\delta))$ yields Eq. (26). $\square$

For the one-dimensional joint extrapolation, take $a_j = b_j$ and

$$\sigma_{j,\text{path}}^2 := \text{Var}(X_{s_j, \lambda(s_j)}).$$

The equal-shot and optimized shot complexities become

$$N_{1\text{D},\text{eq}} = O \left[\left(\frac{m \sum_j b_j^2 \sigma_{j,\text{path}}^2}{\varepsilon^2} + \frac{\|O\| m \max_j |b_j|}{\varepsilon} \right) \log \frac{1}{\delta} \right], \quad (27)$$

and

$$N_{1\text{D},\text{opt}} = O \left[\left(\frac{(\sum_j |b_j| \sigma_{j,\text{path}})^2}{\varepsilon^2} + \frac{\|O\| \|\mathbf{b}\|_1}{\varepsilon} \right) \log \frac{1}{\delta} \right]. \quad (28)$$

For a tensor-product two-dimensional extrapolation with step-size coefficients b_j and noise-direction coefficients d_ℓ , the estimator has coefficients $a_{j,\ell} = b_j d_\ell$. Writing

$$\sigma_{j,\ell}^2 := \text{Var}(X_{s_j, \lambda_\ell}),$$

we similarly get

$$N_{2\text{D},\text{eq}} = O \left[\left(\frac{m_s m_\lambda \sum_{j,\ell} b_j^2 d_\ell^2 \sigma_{j,\ell}^2}{\varepsilon^2} + \frac{\|O\| m_s m_\lambda \max_{j,\ell} |b_j d_\ell|}{\varepsilon} \right) \log \frac{1}{\delta} \right], \quad (29)$$

and

$$N_{2\text{D},\text{opt}} = O \left[\left(\frac{(\sum_{j,\ell} |b_j d_\ell| \sigma_{j,\ell})^2}{\varepsilon^2} + \frac{\|O\| \|\mathbf{b}\|_1 \|\mathbf{d}\|_1}{\varepsilon} \right) \log \frac{1}{\delta} \right]. \quad (30)$$

Thus the Bernstein estimate does not by itself change the worst-case asymptotic scaling in ε , but it replaces the range-only prefactor by the actual variance profile of the measured extrapolation data.

For a single Pauli observable, one shot gives $X \in \{\pm 1\}$ and hence

$$\sigma^2 = \text{Var}(X) = 1 - \langle O \rangle^2, \quad \langle O \rangle = \mathbb{E}X. \quad (31)$$

For the average magnetization $M_Z = N_{\text{tot}}^{-1} \sum_{i=1}^{N_{\text{tot}}} Z_i$, Eq. (31) holds exactly if one uses a random-site Pauli estimator: choose i uniformly at random, measure Z_i , and output $X = z_i \in \{\pm 1\}$. Then $\mathbb{E}X = \langle M_Z \rangle$. If instead each shot measures the full bit string and outputs $Y = N_{\text{tot}}^{-1} \sum_i z_i$, then

$$\text{Var}(Y) = \frac{1}{N_{\text{tot}}^2} \sum_{i,k} \text{Cov}(Z_i, Z_k),$$

so correlations enter and Eq. (31) is not generally an equality. Since $Y \in [-1, 1]$, one still has $\text{Var}(Y) \leq 1 - (\mathbb{E}Y)^2$, and the actual full-bit-string estimator can be treated through the empirical variance profile σ_i^2 in the bounds above.

In the random-site Pauli formulation, a sufficient condition for the one-dimensional path to have no larger Bernstein variance proxy than the two-dimensional grid is

$$\sum_j b_j^2 (1 - \mu_{j,\text{path}}^2) \leq \sum_{j,\ell} b_j^2 d_\ell^2 (1 - \mu_{j,\ell}^2), \quad (32)$$

where $\mu_{j,\text{path}} = \mathbb{E}X_{s_j, \lambda(s_j)}$ and $\mu_{j,\ell} = \mathbb{E}X_{s_j, \lambda_\ell}$. A stronger pointwise sufficient condition is

$$1 - \mu_{j,\text{path}}^2 \leq \sum_\ell d_\ell^2 (1 - \mu_{j,\ell}^2) \quad \text{for every } j. \quad (33)$$

For the optimized allocation, the analogous sufficient condition is

$$\sum_j |b_j| \sqrt{1 - \mu_{j,\text{path}}^2} \leq \sum_{j,\ell} |b_j d_\ell| \sqrt{1 - \mu_{j,\ell}^2}. \quad (34)$$

These conditions formalize the intuition that the coupled path is statistically favorable when its measured points have smaller shot variance, while the two-dimensional grid pays the extra noise-direction coefficient weights.

A. Applications to k -local Hamiltonians

Suppose that $\text{ad}_H = \sum_{\gamma \in [\Gamma]} \mathcal{A}_\gamma$ and $\mathcal{L}_{\text{tot}} := \sum_{v \in [M]} \mathcal{L}_v$ are k -local Lindbladians on a lattice $\Lambda = [N]$. We further assume that ad_H is g_H -extensive and $\mathcal{L}_{\text{tot}}$ is g_L -extensive. Recall the unified representation of $\mathcal{A}_\gamma$ and $\mathcal{L}_v$ in Eq. (10) and Eq. (12):

$$\mathcal{K}_j = \begin{cases} -it a_v \text{ad}_{H_{\gamma_v}} & \text{if } j = 2v - 1, \\ \lambda \mathcal{L}_v & \text{if } j = 2v, \end{cases}$$

$$\tilde{\mathcal{K}}_j := \begin{cases} \text{ad}_{H_{\gamma_v}} & \text{if } j = 2v - 1, \\ \mathcal{L}_v & \text{if } j = 2v. \end{cases},$$

for $v = 1, 2, \dots, M$.

Now we substitute $t = sT, \lambda = c(sT)^{p+1}$ and let $\mathcal{G}_{(q_0)}(s)$ be the effective generator defined in Eq. (11). By Lemma 2, the coefficients of the series expansion of $e^{\mathcal{G}_{(q_0)}(s)T}$ can be bounded as powers of μ_{comm, q_0} defined in Eq. (18). We now provide an upper bound of μ_{comm, q_0} .

Lemma 5. *The parameter μ_{comm, q_0} defined in Eq. (18) satisfies*

$$\mu_{\text{comm}, q_0} \leq q_0 N^{1/(p+1)} (2a_{\max} k \Upsilon g_H + (2ckg_L)^{1/(p+1)}).$$

Proof. For the set S_{q_1, q_2} defined in Eq. (13), we decompose it according to the position of the odd and even indices. For any index set $\mathcal{I} \subseteq [q_1 + q_2]$, we define the subset $S_{\mathcal{I}}$ as

$$S_{\mathcal{I}} = \left\{ (j_1, \dots, j_{q_1+q_2}) \in [2M]^{q_1+q_2} \mid j_k \text{ is odd for } k \in \mathcal{I}, \text{ and } j_k \text{ is even for } k \in [q_1 + q_2] \setminus \mathcal{I} \right\}.$$

We can then express S_{q_1, q_2} as the disjoint union over the collection of all size- q_1 subsets of $[q_1 + q_2]$:

$$S_{q_1, q_2} = \bigsqcup_{\mathcal{I} \in \binom{[q_1+q_2]}{q_1}} S_{\mathcal{I}}.$$

The commutator bounds are

$$\begin{aligned} \alpha_{\text{comm}}^{(q_1, q_2)} &= \sum_{(j_1, \dots, j_{q_1+q_2}) \in S_{q_1, q_2}} \left\| [\tilde{\mathcal{K}}_{j_1}, \tilde{\mathcal{K}}_{j_2}, \dots, \tilde{\mathcal{K}}_{j_{q_1+q_2}}] \right\| \\ &= \sum_{\mathcal{I} \in \binom{[q_1+q_2]}{q_1}} \sum_{(j_1, \dots, j_{q_1+q_2}) \in S_{\mathcal{I}}} \left\| [\tilde{\mathcal{K}}_{j_1}, \tilde{\mathcal{K}}_{j_2}, \dots, \tilde{\mathcal{K}}_{j_{q_1+q_2}}] \right\| \\ &\leq \sum_{\mathcal{I} \in \binom{[q_1+q_2]}{q_1}} N(q_1 + q_2 - 1)! (2k)^{q_1+q_2-1} (\Upsilon g_H)^{q_1} g_L^{q_2} \\ &= \binom{q_1 + q_2}{q_1} N(q_1 + q_2 - 1)! (2k)^{q_1+q_2-1} (\Upsilon g_H)^{q_1} g_L^{q_2} \\ &\leq \binom{q_1 + (p+1)q_2}{q_1} N(q_1 + q_2)! (2k \Upsilon g_H)^{q_1} ((2kg_L)^{1/(p+1)})^{q_2(p+1)}, \end{aligned}$$

where the third line follows from [Corollary 1](#). Then we have

$$\begin{aligned}
\mu_{\text{comm}, q_0} &= \max_{q' \geq p+1} \left(\sum_{\substack{1 \leq q_1 + q_2 \leq q_0 \\ q_1 + (p+1)q_2 = q'}} \frac{a_{\text{max}}^{q_1} c^{q_2}}{(q_1 + q_2)^2} \alpha_{\text{comm}}^{(q_1, q_2)} \right)^{1/q'} \\
&\leq \max_{q' \geq p+1} \left(\sum_{\substack{1 \leq q_1 + q_2 \leq q_0 \\ q_1 + q_2(p+1) = q'}} \binom{q_1 + q_2(p+1)}{q_1} N(q_1 + q_2)! (2a_{\text{max}} k \Upsilon g_H)^{q_1} ((2ckg_L)^{1/(p+1)})^{q_2(p+1)} \right)^{1/q'} \\
&\leq \max_{q' \geq p+1} \left(\sum_{q_1=1}^{q_0} \binom{q'}{q_1} N q_0^{q'} (2a_{\text{max}} k \Upsilon g_H)^{q_1} ((2ckg_L)^{1/(p+1)})^{q' - q_1} \right)^{1/q'} \\
&\leq q_0 N^{1/(p+1)} \max_{q' \geq p+1} ((2a_{\text{max}} k \Upsilon g_H + (2ckg_L)^{1/(p+1)})^{q'})^{1/q'} \\
&= q_0 N^{1/(p+1)} (2a_{\text{max}} k \Upsilon g_H + (2ckg_L)^{1/(p+1)}).
\end{aligned}$$

□

Now we apply [Theorem 2](#) to k -local Hamiltonians. In the following theorem, we make the dependence on locality, extensiveness, and intrinsic noise explicit.

Theorem 3. *Suppose that the intrinsic noise strength of the device, λ_{min} , satisfies*

$$\lambda_{\text{min}} = \tilde{O} \left(\frac{1}{(Nk\Upsilon a_{\text{max}} g_H T)^{1+1/p} k g_L} \right). \quad (35)$$

There exists an algorithm that, given any observable O and initial state ρ_0 , estimates $\text{Tr}[O e^{-\text{iad}_H T} \rho_0]$ to within an additive error of $\epsilon \|O\|$ using a total of

$$O \left(\max \{ \log^3(1/\epsilon), \Xi \log^2(1/\epsilon) \} \frac{(\log \log(1/\epsilon))^3}{\epsilon^2} \right)$$

Trotter steps, where we define the parameter-dependent factor

$$\Xi := N^{1/p} (k\Upsilon a_{\text{max}} g_H T)^{1+1/p} \log^{1+1/p}(N/\epsilon).$$

This involves $\tilde{O}(1/\epsilon^2)$ independent circuit runs, each comprising at most

$$O \left(\max \{ \log^3(1/\epsilon), \Xi \log^2(1/\epsilon) \} \right)$$

Trotter steps.

Proof. Note that $\mathcal{K} := \sum_{j=1}^{2M} \mathcal{K}_j$ is k -local and $(\Upsilon s T a_{\text{max}} g_H + c(sT)^{p+1} g_L)$ -extensive. Following the proof of [Lemma 12](#), the doubly right-nested commutator bound of $\mathcal{K}$, denoted $\bar{\alpha}_{\text{comm}, q_0}(s)$, satisfies

$$\bar{\alpha}_{\text{comm}, q_0}(s) \leq 2(4eq_0 k (\Upsilon s T a_{\text{max}} g_H + c(sT)^{p+1} g_L))^{q_0+1} N,$$

provided $4eq_0 k (\Upsilon s T a_{\text{max}} g_H + c(sT)^{p+1} g_L) \leq 1/2$. Let $m = O(\log(1/\epsilon))$ be the number of extrapolation points. We choose the expansion order as $q_0 = \lceil \log(8e^2 N \log m / \epsilon) \rceil = O(\log(N/\epsilon))$.

We set the step size s_0 as

$$\left[2eq_0 N^{1/(p+1)} T (4ek\Upsilon a_{\text{max}} g_H + (4ekcg_L)^{1/(p+1)}) \right]^{-1-1/p} \quad (36)$$

$$= \Theta \left(\left[\begin{array}{c} \log^{1+1/p}(N/\epsilon) N^{1/p} T^{1+1/p} \\ \times (k\Upsilon a_{\text{max}} g_H + (kcg_L)^{1/(p+1)})^{1+1/p} \end{array} \right]^{-1} \right).$$

One can verify that this choice of s_0 satisfies the condition $4eq_0 k (\Upsilon s_0 T a_{\text{max}} g_H + c(s_0 T)^{p+1} g_L) \leq 1/2$. Furthermore, we have

$$\begin{aligned}
\frac{\bar{\alpha}_{\text{comm}, q_0}(s_0)}{s_0} &\leq 2N \max \left\{ \frac{(s_0^{p/(p+1)} 4eq_0 k \Upsilon T a_{\text{max}} g_H)^{q_0+1}}{(4eq_0 kcg_L s_0^p T^{p+1})^{q_0+1}}, \right\} \\
&\leq 2e^{-q_0-1} N \leq \frac{\epsilon}{8e^2 C \log m}.
\end{aligned}$$

Also, s_0 satisfies the condition related to μ_{comm, q_0} :

$$\begin{aligned}
s_0 &\leq \frac{1}{e(2q_0 N^{1/(p+1)} T (2a_{\text{max}} k \Upsilon g_H + (2ckg_L)^{1/(p+1)}))^{1+1/p}} \\
&\leq \frac{1}{e(2\mu_{\text{comm}, q_0} T)^{1+1/p}}.
\end{aligned}$$

To optimize the constraint on λ_{min} without increasing the asymptotic complexity, we set

$$c = \frac{(k\Upsilon a_{\text{max}} g_H)^{p+1}}{kg_L}, \quad (37)$$

which implies $(kcg_L)^{1/(p+1)} = k\Upsilon a_{\text{max}} g_H$. Substituting this into Eq. (36), the inverse step size scales as:

$$\frac{1}{s_0} = O \left(N^{1/p} (k\Upsilon a_{\text{max}} g_H T)^{1+1/p} \log^{1+1/p}(N/\epsilon) \right) = O(\Xi).$$

Now, applying [Theorem 2](#), the algorithm requires a total number of Trotter steps of:

$$O \left(\max \{ \log^3(1/\epsilon), \Xi \log^2(1/\epsilon) \} \frac{(\log \log(1/\epsilon))^3}{\epsilon^2} \right).$$

The maximum number of Trotter steps per run is

$$O\left(\max\left\{\log^3(1/\varepsilon), \Xi \log^2(1/\varepsilon)\right\}\right).$$

This result holds provided that the intrinsic noise strength satisfies:

$$\lambda_{\min} = \tilde{O}(c(s_0 T)^{p+1}) = \tilde{O}\left(\frac{1}{(Nk\Upsilon a_{\max} g_H T)^{1+1/p} k g_L}\right).$$

□

Refs. [32, 33] established that under Pauli noise, any unbiased QEM protocol requires a sampling overhead of at least $\Omega(1/\varepsilon^2)$ to achieve additive error ε , and that the maximum circuit depth tolerable at a given noise rate is fundamentally limited. Our joint mitigation protocol achieves the optimal $\tilde{O}(1/\varepsilon^2)$ sampling overhead, and the feasibility condition on $\lambda_{\min}$ in Eq. (35) matches the fundamental noise threshold identified in these works. Thus, under the mild-noise assumption, our algorithm operates at the information-theoretic limit of what QEM can achieve without resorting to full quantum error correction. In the noiseless limit ($g_L = 0$), Theorem 3 provides a resource analysis of the Trotter error mitigation algorithm proposed by Watson and Watkins [1] for k -local lattice Hamiltonians. This result parallels that of Mizuta [13], which analyzed the complexity of multi-product formulas without physical noise using a different proof technique.

A crucial implication of Theorem 3 is that, even when the device is subject to local noise satisfying Eq. (35), the noiseless extrapolation algorithm can be applied with only a constant complexity overhead by properly coupling the Trotter step size and the noise strength in the extrapolation. This implies that physical noise mitigation is achieved at virtually no asymptotic cost. Specifically, our choice of c in Eq. (37) merely rescales the step size s_0 by a factor of $2^{-1-1/p}$. Consequently, the total complexity, which is proportional to s_0^{-1} , increases only by a constant factor of $2^{1+1/p}$.

XZ: I'm thinking whether to explicitly compare our joint mitigation algorithm with the nested mitigation one. Since the asymptotic complexity advantage of our algorithm is only $\log(1/\varepsilon)(\log \log(1/\varepsilon))^2$. If not, we can instead claim that our algorithm can mitigate the physical error almost for free while doing Trotter error mitigation.

B. Applications to geometrically-local Hamiltonian

In this section, we consider a d -dimensional lattice $\Lambda \subseteq \mathbb{Z}^d$ and assume that $\text{ad}_H = \sum_{\gamma \in [\Gamma]} \mathcal{A}_\gamma$ is geometrically-local. For any $x \in \Lambda$ and a positive number R , define $B_R(x)$ as the ℓ_1 ball centered on x with radius R :

$$B_R(x) = \{y \in \Lambda : d(x, y) \leq R\},$$

where

$$d(x, y) = \sum_{i=1}^d |x_i - y_i|,$$

is the ℓ_1 distance between $x = (x_1, \dots, x_d)$ and $y = (y_1, \dots, y_d)$. Assume that we can group the summand in the decomposition of ad_H and $\mathcal{L}_{\text{tot}}$ as

$$\text{ad}_H = \sum_{x \in \Lambda} \mathcal{A}_x, \quad \mathcal{A}_x = \sum_{\gamma \in \mathcal{I}_x} \mathcal{A}_\gamma,$$

where $\{\mathcal{I}_x\}_{x \in \Lambda}$ is a partition of $[\Gamma]$, such that

$$\text{supp}(\mathcal{A}_\gamma) \subseteq B_R(x), \quad \forall \gamma \in \mathcal{I}_x,$$

for some positive number R . We also assume that the observable of interest O is supported on $B_{R_O}(x_O)$ for some positive R_O and $x_O \in \Lambda$. Define

$$J_H = \max_{x \in \Lambda} \sum_{\gamma \in \mathcal{I}_x} \|\mathcal{A}_\gamma\|.$$

Define $\Lambda_d = 2^d/d!$ such that

$$|B_R(x)| = \Lambda_d R^d + O(R^{d-1}) \leq C_d R^d, \quad \forall x \in \Lambda,$$

for some $C_d \leq 2\Lambda_d$. Then ad_H and $\mathcal{L}_{\text{tot}}$ are g_H and g_L -extensive with

$$g_H \leq \max_{x \in \Lambda} \sum_{y \in B_R(x)} \sum_{\gamma \in \mathcal{I}_y} \|\mathcal{A}_\gamma\| \leq |B_R(x)| J_H = O(C_d R^d J_H).$$

For any $l \geq 0$, define

$$\text{ad}_{H_{(l)}} = \sum_{y \in B_{R_O+l+R}(x_O)} \mathcal{A}_y,$$

be the truncated Hamiltonian terms within a ball around the support of O . The difference between the time-evolved observables O under $\text{ad}_{H_{(l)}}$ and ad_H can be bounded as follows.

Lemma 6 ([34, Lemma 2]). *For any $l \geq 0$, we have*

$$\begin{aligned} & \left\| e^{\text{iad}_H t} O - e^{\text{iad}_{H_{(l)}} t} O \right\| \\ & \leq |\text{supp}(O)| \|O\| \min\{e^{-l/R}(e^{eC_d R^d t} - 1), 1\}. \end{aligned}$$

We can choose

$$l = eC_d R^{d+1} T + R \log(2|\text{supp}(O)|/\varepsilon),$$

such that for any initial state ρ_0

$$\begin{aligned} & \left| \text{Tr}[O e^{-\text{iad}_H T} \rho_0] - \text{Tr}[O e^{-\text{iad}_{H_{(l)}} T} \rho_0] \right| \\ & \leq \left\| e^{\text{iad}_H T} O - e^{\text{iad}_{H_{(l)}} T} O \right\| \leq \varepsilon/2. \end{aligned}$$

Therefore, we can simulate the truncated Hamiltonian $H_{(l)}$ on $B_{R_0+l+2R}(x_O)$, and the noise terms are restricted to $B_{R_0+l+2R}(x_O)$. Define

$$\mathcal{L}_{(l),\text{tot}} = \sum_{v \in B_{R_0+l+2R}(x_O)} \mathcal{L}_v,$$

which includes all possible noise terms. **XZ: TBD Shuo:** The intended next step is to combine the locality truncation above with **Theorem 3**. Let

$$N_{\text{eff}} = |B_{R_0+l+2R}(x_O)| \\ = O(C_d(R_O + eC_d R^{d+1}T + R \log(|\text{supp}(O)|/\varepsilon) + R)^d)$$

be the effective number of sites in the truncated light cone. Applying **Theorem 3** to the truncated Hamiltonian $H_{(l)}$ and the restricted noise generator $\mathcal{L}_{(l),\text{tot}}$ gives an estimator for $\text{Tr}[Oe^{-i\text{ad}_{H_{(l)}}T} \rho_0]$ with error at most $\varepsilon\|O\|/2$. Together with the truncation error above, this yields an $\varepsilon\|O\|$ -accurate estimator for the original local observable. The resulting complexity should be obtained from **Theorem 3** by replacing the global system size N with N_{eff} , so the dependence on the total lattice size disappears and is replaced by the volume of the effective light cone around $\text{supp}(O)$.

IV. EXPERIMENTAL DEMONSTRATION

Roadmap. This section illustrates the joint extrapolation protocol in two complementary settings. The hardware experiment demonstrates that the learned-channel ZNE stage and the Trotter-step extrapolation stage can be combined on a superconducting device with realistic noise. The tensor-network simulations provide a controlled comparison with a two-dimensional extrapolation baseline and test the same mechanism on larger one-dimensional instances. These demonstrations are intended to support the practical workflow of the protocol; the asymptotic scaling claims are those established by the theoretical bounds above.

The Hamiltonian used in both demonstrations is a one-dimensional transverse-field Ising model,

$$H = -J \sum_{\langle i,j \rangle} Z_i Z_j - h \sum_{i=1}^n X_i,$$

with periodic boundary conditions on a 9-qubit ring. The system is evolved for a total time T using a second-order symmetric (i.e., $p = 1$) Trotter decomposition, **Shuo: numerics to align with the topological structure.**

$$\mathcal{P}(s) = e^{-ihs \sum_i X_i/2} e^{-iJs \sum_{\langle i,j \rangle} Z_i Z_j} e^{-ihs \sum_i X_i/2},$$

where $s = T/r$ is the Trotter step size and r is the number of Trotter steps. In the experimental plots below we also write the Trotter step number as N_{Trotter} when this avoids confusion with the number of qubits.

The observable measured throughout the hardware experiments is the average longitudinal magnetization,

$$O = \frac{1}{n} \sum_{i=1}^n Z_i,$$

which can be estimated directly from computational-basis measurements.

A. Real quantum hardware experiments

We perform the hardware experiments on the 13-qubit Tianji S2 superconducting quantum chip, using a connected 9-qubit subgraph to implement the Trotterized Ising dynamics. The logical circuit is mapped to the device topology as shown in **Figure 1**: each transverse-field layer is implemented by parallel single-qubit rotations, while the Ising interaction layer is scheduled into three disjoint patterns of R_{ZZ} gates. All logical gates are then compiled into the native gate set of the device, with R_X and R_{ZZ} rotations implemented using H , R_Z , and controlled- Z gates.

To characterize the physical noise relevant to these circuits, we adopt the sparse Pauli-Lindblad model of Ref. [20]. In this model, each effective circuit layer is followed by a noise generator

$$\mathcal{L}(\rho) = \sum_{k \in \mathcal{K}} \lambda_k (P_k \rho P_k - \rho). \quad (38)$$

Here $\mathcal{K}$ denotes the retained set of Pauli components, and sparsity means that $|\mathcal{K}| \ll 4^n - 1$ grows only polynomially with the number of qubits. Exponentiating (38) gives the Pauli channel

$$\Lambda(\rho) = \prod_k \left(w_k \cdot + (1 - w_k) P_k \cdot P_k^\dagger \right) \rho,$$

where $w_k = (1 + e^{-2\lambda_k})/2$. **Shuo: In our implementation the Hamiltonian parameters are written equivalently as rotation rates $r_x = -h$ and $r_{zz} = -J$.**

The coefficients λ_k are learned from Clifford training circuits that preserve the same qubit mapping, layer structure, and native-gate compilation as the target Trotter circuits, but restrict the rotation angles to Clifford values. We compare the noiseless Clifford expectations obtained by classical simulation with the corresponding hardware measurements, after readout mitigation, and solve the resulting sparse inverse problem to obtain an effective layerwise Pauli-Lindblad channel. Additional details of the readout calibration, circuit construction, and noise-learning workflow are given in **Appendix C**.

The learned channel is then used to amplify the effective noise strength through random Pauli insertions, enabling zero-noise extrapolation at fixed Trotter step number. Since this ZNE stage removes only the stochastic hardware bias, the remaining deterministic product-formula error is further reduced by extrapolating the mitigated estimates in $x = 1/N_{\text{Trotter}}$. **Figure 2** shows the

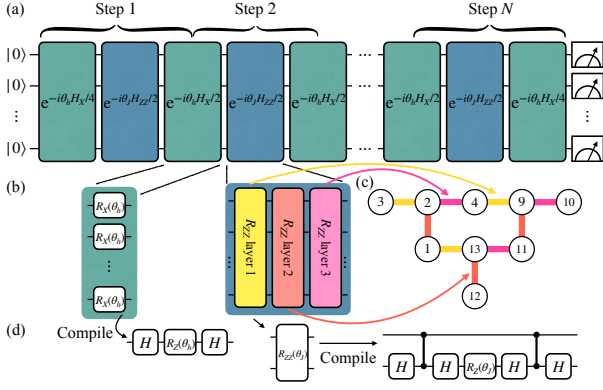


FIG. 1. Hardware implementation of Trotterized Ising dynamics. (a) A second-order symmetric Trotter step alternates transverse-field rotations and Ising interactions. (b) The interaction layer is decomposed into three parallel R_{ZZ} patterns. (c) A connected 9-qubit subgraph of the 13-qubit device is used; colored edges show the scheduled interaction patterns. (d) Logical R_X and R_{ZZ} operations are compiled into the native H , R_Z , and controlled- Z gates.

resulting combined extrapolation for two representative evolution settings. In both cases, the hardware data follow the same trend as the corresponding numerical simulation and extrapolate toward the exact reference value at $x = 0$. This provides an experimental demonstration of workflow compatibility between learned-channel noise mitigation and Trotter-step extrapolation, rather than a standalone verification of the asymptotic scaling bounds.

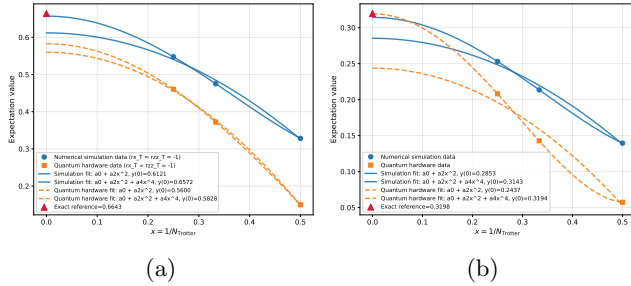


FIG. 2. Combined hardware extrapolation of noise and Trotter errors. Numerical simulation (blue) and superconducting-hardware data (orange) are plotted versus $x = 1/N_{\text{Trotter}}$. The fitted curves extrapolate toward $x = 0$, where both the finite-step Trotter error and the amplified physical-noise bias are removed in the fitted model; the red triangle marks the exact reference. Panels show two representative evolution settings: (a) $r_x T = r_z T = -1$ and (b) $r_x T = r_z T = -2$. The agreement of the hardware and numerical trends supports the composability of learned-channel ZNE with Trotter-step extrapolation on the tested device.

B. Classical tensor network simulation

In addition to the hardware experiment, whose size and depth are limited by device fidelity, we use classical tensor-network simulations to study larger one-dimensional Ising instances and to compare extrapolation strategies in a controlled setting. The results and plots in this subsection are obtained by numerical simulations on a local machine with an Apple M2 CPU (8 cores) and 16 GB unified memory via Python 3.11.12. We use the ‘quimb’ library [35] to represent (noisy) quantum circuits using tensor networks. The target observable is again the average longitudinal magnetization. We compare three mitigation strategies: the proposed joint one-dimensional extrapolation, a noise-free Trotter extrapolation that removes only the algorithmic Trotter error, and a two-dimensional extrapolation baseline that treats the Trotter step size and the physical noise strength as independent extrapolation variables. The goal of this comparison is to test whether coupling the physical noise strength to the Trotter step size can retain the accuracy of the two-dimensional strategy while using a one-dimensional extrapolation procedure at the scale of dozens of qubits. For the two-dimensional baseline, we use a tensor-product Richardson extrapolation. The Trotter direction uses the four step counts $r \in \{1, 2, 3, 7\}$, equivalently step sizes $1/r$, and the noise direction uses the four positive noise-amplification factors $\{49, 9, 4, 1\}$, extrapolated to zero noise. **Shuo: In practice, explain why use square instead of cubic, mainly to align with the experiments** Thus, the Richardson order in each direction is determined by the number of data points in that direction, and the two-dimensional baseline uses $4 \times 4 = 16$ noisy data points. The noiseless curve labeled “Noise 0” is included as a reference and is not counted as a noisy ZNE data point.

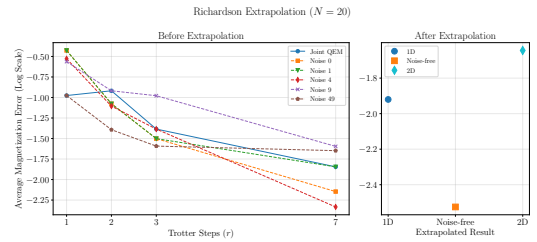


FIG. 3. Average magnetization error as a function of the number of Trotter steps r for a 1D transverse-field Ising model with $n = 20$ qubits. The curves compare the raw noisy estimates at several noise-amplification levels with the estimates obtained by the joint one-dimensional extrapolation, the noise-free Trotter extrapolation, and the tensor-product two-dimensional Richardson baseline. The joint one-dimensional extrapolation tracks the accuracy of the two-dimensional baseline while using a single extrapolation variable. Results are obtained by classical tensor-network simulation with quimb at seed 43.

It can be seen from Figure 3 and Figure 4 that the

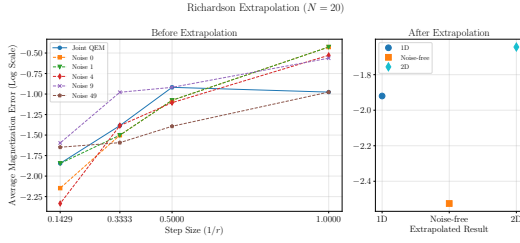


FIG. 4. The same average magnetization error data plotted against the Trotter step size $1/r$. This view highlights the extrapolation behavior as the step size approaches zero. The inset compares the final extrapolated estimates from the joint one-dimensional method, the noise-free Trotter extrapolation, and the two-dimensional baseline, showing that coupling the noise strength to the step size can reproduce the relevant two-dimensional correction in this controlled instance. Results are obtained by classical tensor-network simulation with `quimb` at seed 43.

proposed one-dimensional joint extrapolation substantially reduces the bias of the noisy estimates across the simulated Trotter step sizes. In the displayed instance, the final extrapolated value obtained by the one-dimensional method achieves comparable accuracy to the two-dimensional Richardson baseline while using a one-dimensional extrapolation procedure. This supports the central intuition of the protocol: once the physical noise strength is coupled to the Trotter step size at the appropriate order, a single extrapolation variable can remove the leading physical-noise and Trotter contributions simultaneously.

V. CONCLUSION AND DISCUSSION

We have developed a commutator-sensitive analysis of Trotter extrapolation in the presence of physical noise. Our analysis builds on the finite-order BCH truncation bound from Ref. [14], which is controlled by doubly right-nested commutators and avoids requiring convergence of the full BCH series. In the present work, we use this tool to analyze noisy product formulas and to extend the commutator-scaling guarantees of Trotter error mitigation to a joint mitigation protocol for algorithmic and physical errors.

The main contribution of this paper is a one-dimensional joint extrapolation protocol for simultaneously mitigating algorithmic Trotter error and physical noise. By coupling the noise strength to the Trotter step size as $\lambda(s) = c(sT)^{p+1}$ for a p -th order product formula, the leading Trotter and noise contributions enter at the same order in the extrapolation variable. Richardson extrapolation can therefore remove both contributions using a single set of extrapolation nodes. For k -local and geometrically-local Hamiltonians under sufficiently mild local noise, our bounds show that physical noise can be mitigated together with Trotter error with only a

constant asymptotic overhead compared with the corresponding noiseless Trotter extrapolation protocol.

Several assumptions are important for interpreting these results. The protocol requires access to a tunable family of effective noise strengths, at least in the range needed by the extrapolation nodes. In practice, this tuning may be implemented by noise amplification techniques such as unitary folding, or by model-dependent procedures based on a learned sparse Pauli-Lindblad noise model. The achievable range and accuracy of this tuning depend on the hardware platform and on the quality of noise characterization. Our theoretical analysis also focuses on local Lindbladian noise and on expectation-value estimation; extending the same framework to more general non-Markovian noise models, correlated readout errors, or adaptive measurement strategies remains an interesting direction.

The numerical and hardware demonstrations support the basic mechanism of the protocol, but they are not intended to exhaust the possible performance regimes. Future work could optimize the choice of extrapolation nodes and the coupling constant c , quantify constant-factor overheads for concrete devices, and test the protocol on larger Hamiltonian simulation tasks where Trotter error and physical noise are both visible at experimentally accessible circuit depths.

Appendix A: Preliminaries

1. Richardson extrapolation

Lemma 7 (see [36, Lemma A.9]). *Suppose that function $f: \mathbb{R} \rightarrow \mathbb{R}$ can be decomposed as*

$$f(x) = P_m(x) + R_m(x),$$

where $P_m(x)$ is a degree- $(m-1)$ polynomial and $R_m(x)$ is the remainder. Given any $t > 0$ and $s \in (0, 1)$, we can choose

$$K = \max \left\{ \frac{m}{\pi}, \frac{2t}{s} \right\}, \quad r_j = \left\lceil \frac{K}{\sin^2(\pi(2j-1)/8m)} \right\rceil,$$

and

$$s_j = \frac{t}{r_j}, \quad b_j = \prod_{\ell \neq j} \frac{1}{1 - r_\ell/r_j}$$

so that $1 \leq \|\mathbf{b}\|_1 \leq C \log m$ for some absolute constant C , $r_j = O(\max\{m^3, m^2 t/s\}/j^2)$, and $\max_j \{s_j\} \leq s$. Then define the m -term Richardson extrapolation

$$F^{(m)}(s) = \sum_{j=1}^m b_j f(s_j), \quad (\text{A1})$$

which satisfies

$$|F^{(m)}(s) - f(0)| \leq \|\mathbf{b}\|_1 \max_j |R_m(s_j)|. \quad (\text{A2})$$

2. Technical lemmas

Let $\mathcal{A}(\tau)$ and $\mathcal{B}(\tau)$ be two continuous operator-valued functions. We rely on the following lemma to decompose the time-ordered exponential of the sum $\mathcal{A}(\tau) + \mathcal{B}(\tau)$. This result is essentially the interaction picture representation of the evolution.

Lemma 8 ([37, Page 21]). *Let $\mathcal{H}(\tau) = \mathcal{A}(\tau) + \mathcal{B}(\tau)$ be an operator-valued function defined for $\tau \in \mathbb{R}$ with continuous summands $\mathcal{A}(\tau)$ and $\mathcal{B}(\tau)$. Then*

$$\begin{aligned} \exp_{\mathcal{T}} \left(\int_0^t \mathcal{H}(\tau) d\tau \right) &= \exp_{\mathcal{T}} \left(\int_0^t \mathcal{A}(\tau) d\tau \right) \\ &\quad \cdot \exp_{\mathcal{T}} \left(\int_0^t \exp_{\mathcal{T}}^{-1} \left(\int_0^{\tau_1} \mathcal{A}(\tau_2) d\tau_2 \right) \mathcal{B}(\tau_1) \exp_{\mathcal{T}} \left(\int_0^{\tau_1} \mathcal{A}(\tau_2) d\tau_2 \right) d\tau_1 \right). \end{aligned}$$

We use the following lemma to bound the difference between powers of a superoperator and a quantum channel.

Lemma 9 ([14, Lemma 3]). *Let $\mathcal{N}_1, \mathcal{N}_2$ be two superoperators, where $\mathcal{N}_1$ is a quantum channel. For any positive integer k , the difference between $\mathcal{N}_1^k$ and $\mathcal{N}_2^k$ can be bounded as*

$$\|\mathcal{N}_1^k - \mathcal{N}_2^k\| \leq k \max\{1, \|\mathcal{N}_2\|\}^{k-1} \|\mathcal{N}_1 - \mathcal{N}_2\|.$$

Appendix B: BCH Truncation Error of Different Hamiltonians

Shuo: Consider move this applications to Sec 3. or appendix

In this section, we demonstrate the power of **Theorem 1** in **Section II C** by applying it to the k -local and the second-quantized electronic Hamiltonians.

1. Application to k -local systems

We now discuss the k -local systems that motivated our analysis. Suppose that each generator $\mathcal{K}_v$ is a k -local superoperator on a lattice $\Lambda = [N]$. Formally, this means

$$\mathcal{K}_v = \sum_{\gamma=1}^{\Gamma} \mathcal{K}_{\gamma}^{(v)}, \quad \text{with } |\text{supp}(\mathcal{K}_{\gamma}^{(v)})| \leq k, \quad \forall v \in [M], \gamma \in [\Gamma].$$

Note that while our motivation comes from Lindbladians, the following derivation relies solely on the locality structure and holds for general superoperators. For any site $j \in \Lambda$, we further suppose that

$$\sum_{v=1}^M \sum_{\gamma \in [\Gamma]: j \in \text{supp}(\mathcal{K}_{\gamma}^{(v)})} \|\mathcal{K}_{\gamma}^{(v)}\| \leq g,$$

which is called g -extensiveness of $\mathcal{K} := \sum_{v=1}^M \mathcal{K}_v$.

A wide class of physically relevant models, specifically Lindbladians generated by sparse Pauli strings, naturally fits into this framework. In the following lemma, we translate the Hamiltonian and jump operator parameters into the locality and extensiveness of the resulting superoperators.

Lemma 10. Consider a Lindbladian $\mathcal{L} = \mathcal{H} + \sum_{v=1}^M \mathcal{D}[D_v]$ acting on N qubits, where

1. the Hamiltonian $H = \sum_{\gamma} h_{\gamma} P_{\gamma}$ consists of Pauli strings with weight at most k ; and
2. each jump operator $D_v = \sum_{\eta} d_{v,\eta} Q_{v,\eta}$ consists of Pauli strings with weight at most k .

Then the total Lindbladian $\mathcal{L}$ can be decomposed as a sum of $2k$ -local superoperators:

$$\mathcal{L} = \sum_{\mu} \mathcal{K}_{\mu}, \quad \text{with } |\text{supp}(\mathcal{K}_{\mu})| \leq 2k.$$

Furthermore, let the Hamiltonian extensiveness be $g_H = \max_j \sum_{\gamma: j \in \text{supp}(P_{\gamma})} |h_{\gamma}|$ and the dissipative extensiveness be $g_D = \max_j \sum_v \sum_{\eta, \lambda: j \in \text{supp}(Q_{v,\eta}) \cup \text{supp}(Q_{v,\lambda})} |d_{v,\eta} d_{v,\lambda}^*|$. The system satisfies the G -extensiveness condition with parameter

$$G = 2g_H + 2g_D.$$

Proof. The Hamiltonian part $\mathcal{H} = -i \sum_{\gamma} h_{\gamma} [P_{\gamma}, \cdot]$ is a sum of terms with locality $|\text{supp}(P_{\gamma})| \leq k$. Since $\|[P_{\gamma}, \cdot]\| \leq 2\|P_{\gamma}\| = 2$, each term contributes at most $2|h_{\gamma}|$ to the local extensiveness.

For the dissipative part, we expand $\mathcal{D}[D_v](\rho) = \sum_{\eta, \lambda} d_{v,\eta} d_{v,\lambda}^* \mathcal{S}_{\eta, \lambda}(\rho)$, where the basis superoperator is $\mathcal{S}_{\eta, \lambda}(\rho) = Q_{v,\eta} \rho Q_{v,\lambda}^{\dagger} - \frac{1}{2} \{Q_{v,\lambda}^{\dagger} Q_{v,\eta}, \rho\}$. Each $\mathcal{S}_{\eta, \lambda}$ acts non-trivially only on $\text{supp}(Q_{v,\eta}) \cup \text{supp}(Q_{v,\lambda})$, so its locality is bounded by $2k$. Regarding the norm, since $Q_{v,\eta}$ and $Q_{v,\lambda}$ are Pauli strings with unit operator norm, standard norm inequalities for unitary conjugation and anti-commutators imply $\|\mathcal{S}_{\eta, \lambda}\| \leq 1 + \frac{1}{2}(2) = 2$. Summing the coefficients weighted by this norm bound directly yields the extensiveness parameter $G = 2g_H + 2g_D$. $\square$

This lemma ensures that the results we derive for general local superoperators are directly applicable to sparse Pauli-Lindbladians by simply substituting the locality $k \rightarrow 2k$ and extensiveness $g \rightarrow G$.

For general k -local Hamiltonians, the standard commutator bound defined in Eq. (5) is $\alpha_{\text{comm}}^{(q)} = O(q!(kg)^q)$ [13, 31], which causes the BCH series to diverge [13]. This scaling of $\alpha_{\text{comm}}^{(q)}$ remains the same for k -local superoperators, as the key requirement of the proof is locality rather than Hermiticity. Mizuta [13] provided an upper bound on the sum of nested commutators for a single Hamiltonian. However, the proof relies essentially on the commutativity structure determined by the supports of the operators, rather than the specific forms. Therefore, this result admits a straightforward generalization to a sequence of distinct local superoperators.

Lemma 11 (Generalization of [13, Lemma 7]). Let $\mathcal{K}_1, \dots, \mathcal{K}_q$ be a sequence of superoperators on a size- N

lattice Λ such that $\mathcal{K}_j = \sum_{v=1}^M \mathcal{K}_{j,v}$ is k -local and g_j -extensive for all $j \in [q]$. Then the following inequality on the right-nested commutators holds:

$$\begin{aligned} & \sum_{v_1, \dots, v_q=1}^M \left\| [\mathcal{K}_{1,v_1}, \dots, \mathcal{K}_{q',v_{q'}}, \mathcal{A}, \mathcal{K}_{q'+1,v_{q'+1}}, \dots, \mathcal{K}_{q,v_q}] \right\| \\ & \leq q!(2k)^q \left(\prod_{j=1}^q g_j \right) \|\mathcal{A}\|, \end{aligned}$$

for any $q' \in [q]$.

A useful corollary can be obtained by replacing $\mathcal{A}$ in Lemma 11 with a term from a k -local superoperator.

Corollary 1. Let $\mathcal{K}_1, \dots, \mathcal{K}_q$ be a sequence of superoperators on a size- N lattice Λ such that $\mathcal{K}_j = \sum_{v=1}^M \mathcal{K}_{j,v}$ is k -local and g_j -extensive for all $j \in [q]$. Then we have

$$\sum_{v_1, \dots, v_q=1}^M \left\| [\mathcal{K}_{1,v_1}, \dots, \mathcal{K}_{q,v_q}] \right\| \leq N(q-1)!(2k)^{q-1} \prod_{j=1}^q g_j,$$

Proof. We apply Lemma 11 to the superoperator sequence of length $q-1$, $\mathcal{K}_1, \dots, \mathcal{K}_{q-1}$, and set the fixed operator $\mathcal{A} = \mathcal{K}_{q,v_q}$. Summing over the index v_q , we obtain

$$\begin{aligned} & \sum_{v_1, \dots, v_q=1}^M \left\| [\mathcal{K}_{1,v_1}, \dots, \mathcal{K}_{q,v_q}] \right\| \\ & \leq \sum_{v_q=1}^M (q-1)!(2k)^{q-1} \left(\prod_{j=1}^{q-1} g_j \right) \|\mathcal{K}_{q,v_q}\| \\ & = (q-1)!(2k)^{q-1} \left(\prod_{j=1}^{q-1} g_j \right) \sum_{v_q=1}^M \|\mathcal{K}_{q,v_q}\| \\ & \leq (q-1)! \left(\prod_{j=1}^{q-1} 2kg_j \right) N g_q \\ & = N(q-1)!(2k)^{q-1} \prod_{j=1}^q g_j, \end{aligned}$$

where the last inequality uses the property that $\mathcal{K}_q$ is g_q -extensive. $\square$

We can now bound the doubly right-nested commutators defined in Eq. (8) using Lemma 11. In the following theorem, we turn the locality and extensiveness assumptions into the commutator estimate needed by Theorem 1.

Theorem 4. Let $\mathcal{K} = \sum_{v=1}^M \mathcal{K}_v$ be a k -local and g -extensive superoperator on a lattice Λ , and let $\mathcal{A}$ be a superoperator with $|\text{supp}(\mathcal{A})| \leq k$. For a given sequence of positive integers $q_1, q_2, \dots, q_d$, let $P_r = 1 + \sum_{j=r}^d q_j$ for $r = 1, 2, \dots, d$ and $P_{d+1} = 1$. Then the following inequality holds for any position index $q' \in [q_d]$

$$\sum_{v_{1,1}, \dots, v_{d,q_d}=1}^M \left\| \left[[\mathcal{K}_{v_{1,1}}, \dots, \mathcal{K}_{v_{1,q_1}}], \dots, [\mathcal{K}_{v_{d-1,1}}, \dots, \mathcal{K}_{v_{d-1,q_{d-1}}}], \right. \right. \\ \left. \left. [\mathcal{K}_{v_{d,1}}, \dots, \mathcal{K}_{v_{d,q'}}, \mathcal{A}, \mathcal{K}_{v_{d,q'+1}}, \dots, \mathcal{K}_{v_{d,q_d}}] \right] \right\| \leq \left(\prod_{r=1}^d P_{r+1} q_r! (2kg)^{q_r} \right) \|\mathcal{A}\|. \quad (\text{B1})$$

Proof. We first decompose each generator $\mathcal{K}_v$ into a sum of k -local terms. Specifically, let

$$\mathcal{K}_v = \sum_{\gamma=1}^{\Gamma} \mathcal{K}_{\gamma}^{(v)},$$

where each $\mathcal{K}_{\gamma}^{(v)}$ acts non-trivially on at most k sites. To simplify the sum, we re-index these terms as a single sequence $\hat{\mathcal{K}}_j$ for $j = 1, \dots, M\Gamma$, such that

$$\mathcal{K} = \sum_{v=1}^M \mathcal{K}_v = \sum_{j=1}^{M\Gamma} \hat{\mathcal{K}}_j.$$

By the linearity of the commutator and the triangle inequality, the left-hand side of Eq. (B1) is bounded by the sum of the norms of the commutators of the constituent k -local terms

$$\sum_{v_{1,1}, \dots, v_{d,q_d}=1}^M \left\| \left[[\mathcal{K}_{v_{1,1}}, \dots, \mathcal{K}_{v_{1,q_1}}], \dots, [\mathcal{K}_{v_{d,1}}, \dots, \mathcal{K}_{v_{d,q'}}, \mathcal{A}_X, \dots, \mathcal{K}_{v_{d,q_d}}] \right] \right\| \\ \leq \sum_{u_{1,1}, \dots, u_{d,q_d}=1}^{M\Gamma} \left\| \left[[\hat{\mathcal{K}}_{u_{1,1}}, \dots, \hat{\mathcal{K}}_{u_{1,q_1}}], \dots, [\hat{\mathcal{K}}_{u_{d,1}}, \dots, \hat{\mathcal{K}}_{u_{d,q'}}, \mathcal{A}, \dots, \hat{\mathcal{K}}_{u_{d,q_d}}] \right] \right\|. \quad (\text{B2})$$

For any layer $r \in [d]$, let $\vec{u}_r = (u_{r,1}, \dots, u_{r,q_r})$ denote the vector of indices ranging from 1 to $M\Gamma$. We define the nested commutator for layer r as

$$\hat{\mathcal{K}}_{\vec{u}_r} := [\hat{\mathcal{K}}_{u_{r,1}}, \dots, \hat{\mathcal{K}}_{u_{r,q_r}}].$$

Similarly, let $\hat{\mathcal{W}}_{\vec{u}_d} := [\hat{\mathcal{K}}_{u_{d,1}}, \dots, \mathcal{A}_X, \dots, \hat{\mathcal{K}}_{u_{d,q_d}}]$ denote the innermost term containing $\mathcal{A}$ at the q' -th position. The RHS of Eq. (B2) can be rewritten as

$$\sum_{\vec{u}_1, \dots, \vec{u}_d} \left\| [\hat{\mathcal{K}}_{\vec{u}_1}, \dots, \hat{\mathcal{K}}_{\vec{u}_{d-1}}, \hat{\mathcal{W}}_{\vec{u}_d}] \right\|.$$

Define the partial nested commutator starting from layer r as

$$\mathcal{W}_{\vec{u}_r, \dots, \vec{u}_d} := [\hat{\mathcal{K}}_{\vec{u}_r}, \dots, \hat{\mathcal{K}}_{\vec{u}_{d-1}}, \hat{\mathcal{W}}_{\vec{u}_d}].$$

We prove the following bound by backward induction on r from d to 1

$$\sum_{\vec{u}_r, \dots, \vec{u}_d} \left\| \mathcal{W}_{\vec{u}_r, \dots, \vec{u}_d} \right\| \leq \left(\prod_{j=r}^d P_{j+1} q_j! (2kg)^{q_j} \right) \|\mathcal{A}\|. \quad (\text{B3})$$

Consider the base case $r = d$. The term is $\sum_{\vec{u}_d} \|\hat{\mathcal{W}}_{\vec{u}_d}\|$. Since $\mathcal{W}_{\vec{u}_d}$ involves one fixed operator $\mathcal{A}$ and q_d operators from a k -local set, we apply [Lemma 11](#). This yields the bound $q_d! (2kg)^{q_d} \|\mathcal{A}\|$. Since $P_{d+1} = 1$, the base case holds.

Assume Eq. (B3) holds for r . We consider the sum for $r - 1$

$$\sum_{\vec{u}_{r-1}, \dots, \vec{u}_d} \left\| \mathcal{W}_{\vec{u}_{r-1}, \dots, \vec{u}_d} \right\| = \sum_{\vec{u}_r, \dots, \vec{u}_d} \sum_{\vec{u}_{r-1}} \left\| [\hat{\mathcal{K}}_{\vec{u}_{r-1}}, \mathcal{W}_{\vec{u}_r, \dots, \vec{u}_d}] \right\|.$$

Let $X_{\mathcal{W}} = \text{supp}(\mathcal{W}_{\vec{u}_r, \dots, \vec{u}_d})$. Its size is bounded by $|X_{\mathcal{W}}| \leq kP_r$. The inner commutator is nonzero only if at least one operator in $\hat{\mathcal{K}}_{\vec{u}_{r-1}}$ overlaps with $X_{\mathcal{W}}$. We explicitly expand the sum over $\vec{u}_{r-1}$ by iterating over which operator overlaps with $X_{\mathcal{W}}$. Let $j \in [q_{r-1}]$ be the index of this operator. We can upper bound the sum by

$$\begin{aligned}
& \sum_{\vec{u}_{r-1}} \left\| \left[\hat{\mathcal{K}}_{\vec{u}_{r-1}}, \mathcal{W}_{\vec{u}_r, \dots, \vec{u}_d} \right] \right\| \\
& \leq \sum_{j=1}^{q_{r-1}} \sum_{\substack{u_{r-1,j} \in [M\Gamma] \\ \text{supp}(\hat{\mathcal{K}}_{u_{r-1,j}}) \cap X_{\mathcal{W}} \neq \emptyset}} \sum_{\substack{u_{r-1,k} \in [M\Gamma] \\ (\forall k \neq j)}} \left\| \left[\hat{\mathcal{K}}_{u_{r-1,1}}, \dots, \hat{\mathcal{K}}_{u_{r-1,q_{r-1}}}, \mathcal{W}_{\vec{u}_r, \dots, \vec{u}_d} \right] \right\| \\
& \leq \sum_{j=1}^{q_{r-1}} \sum_{\substack{u_{r-1,j} \in [M\Gamma] \\ \text{supp}(\hat{\mathcal{K}}_{u_{r-1,j}}) \cap X_{\mathcal{W}} \neq \emptyset}} (q_{r-1} - 1)! (2kg)^{q_{r-1}-1} \|\hat{\mathcal{K}}_{u_{r-1,j}}\| \cdot 2 \left\| \mathcal{W}_{\vec{u}_r, \dots, \vec{u}_d} \right\| \\
& \leq \sum_{j=1}^{q_{r-1}} (kP_r g) \cdot (q_{r-1} - 1)! (2kg)^{q_{r-1}-1} \cdot 2 \left\| \mathcal{W}_{\vec{u}_r, \dots, \vec{u}_d} \right\| \\
& = q_{r-1} \cdot (kP_r g) \cdot (q_{r-1} - 1)! (2kg)^{q_{r-1}-1} \cdot 2 \left\| \mathcal{W}_{\vec{u}_r, \dots, \vec{u}_d} \right\| \\
& = P_r \cdot q_{r-1}! \cdot (2kg)^{q_{r-1}} \left\| \mathcal{W}_{\vec{u}_r, \dots, \vec{u}_d} \right\|.
\end{aligned}$$

Here, the second inequality follows from applying [Lemma 11](#) to the sum over the $q_{r-1} - 1$ operators not at index j with $\mathcal{A} := \hat{\mathcal{K}}_{u_{r-1,j}}$. The third inequality uses the g -extensiveness property: summing the norm of $\hat{\mathcal{K}}_{u_{r-1,j}}$ over all operators overlapping with $X_{\mathcal{W}}$ yields at most $|X_{\mathcal{W}}|g \leq kP_r g$.

Substituting this back and using the inductive hypothesis leads to

$$\begin{aligned}
& \sum_{\vec{u}_{r-1}, \dots, \vec{u}_d} \left\| \mathcal{W}_{\vec{u}_{r-1}, \dots, \vec{u}_d} \right\| \\
& \leq P_r q_{r-1}! (2kg)^{q_{r-1}} \sum_{\vec{u}_r, \dots, \vec{u}_d} \left\| \mathcal{W}_{\vec{u}_r, \dots, \vec{u}_d} \right\| \\
& \leq P_r q_{r-1}! (2kg)^{q_{r-1}} \left(\prod_{j=r}^d P_{j+1} q_j! (2kg)^{q_j} \right) \|\mathcal{A}\|.
\end{aligned}$$

This completes the induction. $\square$

Plugging this bound into [Theorem 1](#), we can bound the truncation error for a k -local system as follows. Note

that if all $\mathcal{K}_v$ are Hamiltonians, this bound matches the one in Mizuta [[13](#), Theorem 3]. However, we use a distinct proof that applies to general superoperators beyond lattice systems. The following lemma records the resulting truncation guarantee in a form that will be used in the extrapolation analysis.

Lemma 12. *Given k -local and g -extensive superoperators $\mathcal{K}_v$, for any $\varepsilon \in (0, 1/2)$ and $q_0 \geq \log(2N/\varepsilon)$, if $4e^2 kg q_0 \leq 1$, then we have*

$$\left\| \exp \left(\sum_{q=1}^{q_0} \Phi_q \right) - \prod_{v=1}^M e^{\mathcal{K}_v} \right\| \leq \varepsilon.$$

Proof. For any positive integer sequence $q_1, \dots, q_{d+1}$, let

$$P_i = \sum_{j=i}^{d+1} q_j,$$

for $i \leq d+1$, $P_{d+2} = 1$, and $q = q_1 + \dots + q_{d+1}$. Then, we have

$$\begin{aligned}
& \bar{\alpha}_{\text{comm}}^{(q_1, \dots, q_{d+1})} \\
& = \sum_{v_{d+1}, q_{d+1}=1}^M \sum_{v_1, 1, \dots, v_{d+1}, q_{d+1}=1}^M \left\| \left[[\mathcal{K}_{v_1, 1}, \dots, \mathcal{K}_{v_1, q_1}], \dots, [\mathcal{K}_{v_{d+1}, 1}, \dots, \mathcal{K}_{v_{d+1}, q_{d+1}}] \right] \right\| \\
& \leq \left(\prod_{i=1}^{d+1} P_{i+1} q_i! (2kg)^{q_i} \right) \frac{1}{g} \sum_{v_{d+1}, q_{d+1}=1}^M \left\| \mathcal{K}_{v_{d+1}, q_{d+1}} \right\| \\
& = \left(\prod_{i=1}^{d+1} P_{i+1} q_i! (2kg)^{q_i} \right) N,
\end{aligned}$$

where the third line follows from [Theorem 4](#), and the fourth line follows from

$$\sum_{v_{d+1}, q_{d+1}=1}^M \|\mathcal{K}_{v_{d+1}, q_{d+1}}\| \leq \sum_{x \in \Lambda} \sum_{\substack{v=1, \dots, M \\ \text{supp}(\mathcal{K}_v) \ni x}} \|\mathcal{K}_v\| \leq gN.$$

For $q_1, \dots, q_{d+1} \leq q_0$, we have

$$\bar{\alpha}_{\text{comm}}^{(q_1, \dots, q_{d+1})} \leq q^d \left(\prod_{i=1}^{d+1} q_0^{q_i} (2kg)^{q_i} \right) N = q^d (2q_0 kg)^q N,$$

where the inequality follows from $P_{i+1} \leq q$ (noting that $P_{d+2} = 1$ does not contribute a factor of q). Then, we have

$$\begin{aligned} \bar{\alpha}_{\text{comm}, q_0} &\leq (2q_0 kg)^{q_0+1} N + \sum_{r=0}^{\infty} \frac{q^r}{(r+1)!} \sum_{\substack{q=q_0+2 \\ 1 \leq p_1, \dots, p_{r+1} \leq q_0 \\ p_1 + \dots + p_{r+1} = q}}^{\infty} \sum (2q_0 kg)^q N \\ &\leq (2q_0 kg)^{q_0+1} N + \sum_{q=q_0+2}^{\infty} \sum_{\substack{1 \leq p_1, \dots, p_{r+1} \leq q_0 \\ p_1 + \dots + p_{r+1} = q}} (2eq_0 kg)^q N \\ &\leq (2q_0 kg)^{q_0+1} N + \sum_{q=q_0+2}^{\infty} (4eq_0 kg)^q N \\ &\leq (2q_0 kg)^{q_0+1} N + (4eq_0 kg)^{q_0+1} N \sum_{q=1}^{\infty} 2^{-q} \\ &\leq 2(4eq_0 kg)^{q_0+1} N \leq (4e^2 q_0 kg)^{q_0+1} \leq 1, \end{aligned}$$

where the third line follows from

$$\sum_{\substack{1 \leq q_1, \dots, q_{r+1} \leq q_0 \\ q_1 + \dots + q_{r+1} = q}} 1 \leq \sum_{\substack{1 \leq q_1, \dots, q_{r+1} \\ q_1 + \dots + q_{r+1} = q}} 1 = \binom{q-1}{r} \leq 2^q,$$

and the fourth line follows from $e^{-q_0-1} \leq \varepsilon/(2N) \leq 1/(2N)$. As $\bar{\alpha}_{\text{comm}, q_0} \leq 1$ is satisfied, we can apply [Theorem 1](#) to obtain

$$\begin{aligned} \left\| \exp \left(\sum_{q=1}^{q_0} \Phi_q \right) - \prod_{v=1}^M e^{\mathcal{K}_v} \right\| &\leq e \bar{\alpha}_{\text{comm}, q_0} \leq 2e(4eq_0 kg)^{q_0+1} N \\ &\leq 2e^{-q_0} N \leq \varepsilon. \end{aligned}$$

□

Based on the convergence condition in Eq. (6) and the bound of α_{comm} in Eq. (7), the BCH series $\sum_{q=1}^{\infty} \Phi_q$ is guaranteed to converge only when $\sum_{v=1}^M \|\mathcal{K}_v\| = O(1)$, which requires $g = O(1/N)$. In contrast, [Lemma 12](#) shows that the exponential of the truncated BCH formula approximates the exponential product in a much larger regime of $g = O(1/(k \log N))$, where the BCH series is not guaranteed to converge.

2. Application to second-quantized electronic structure

Following the notation of Childs et al. [31], we consider the second-quantized electronic structure Hamiltonian in the plane-wave dual basis. In this representation, the Hamiltonian for a system with n spin-orbitals takes the form

$$H = \underbrace{\sum_{j,k=1}^n t_{j,k} A_j^\dagger A_k}_T + \underbrace{\sum_{j=1}^n u_j N_j}_U + \underbrace{\sum_{j,k=1}^n g_{j,k} N_j N_k}_V, \quad (\text{B4})$$

where $A_j^\dagger, A_j$ are the creation and annihilation operators satisfying $\{A_j, A_k^\dagger\} = \delta_{jk}$, and $N_j = A_j^\dagger A_j$. Here, T represents the kinetic energy with hopping amplitudes $t_{j,k}$, U denotes the single-particle potential with coefficients u_j , and V describes the two-body interaction with strength $g_{j,k}$. Note that $\|A_j^\dagger\| = \|A_k\| = \|N_j\| = 1$. We can apply the triangle inequality to upper bound the norms $\|T\|$, $\|U\|$, and $\|V\|$ by the 1-norm of the corresponding coefficient vectors $\|\vec{t}\|_1$, $\|\vec{u}\|_1$, and $\|\vec{g}\|_1$. Childs et al. [31] proved the follow upper bounds for the 1-norm and ∞ -norm of the coefficient vectors.

Lemma 13 ([31, Proposition F.2]). *The 1-norm and ∞ -norm of the coefficients vectors of the kinetic operator T and potential operators U, V in Eq. (B4) can be bounded*

as

$$\begin{aligned}\|\vec{t}\|_\infty &= O(1/n), & \|\vec{t}\|_1 &= O(n) \\ \|\vec{u}\|_\infty &= O(n), & \|\vec{u}\|_1 &= O(n^2) \\ \|\vec{g}\|_\infty &= O(1), & \|\vec{g}\|_1 &= O(n^2).\end{aligned}$$

Denote the summand in Eq. (B4) by H_v , which gives

$$H = \sum_{v=1}^M H_v,$$

with $M = 2n^2 + n$. The right-nested commutators of H_v can be expressed as

$$W = \sum_{\vec{j}, \vec{k}, \vec{\ell}} w_{\vec{j}, \vec{k}, \vec{\ell}} \cdots (A_{j_x}^\dagger A_{k_x}) \cdots (N_{\ell_y}) \cdots,$$

where $\vec{j}, \vec{k}, \vec{\ell}$ are vectors of orbitals. The largest number of $A_{j_x}^\dagger A_{k_x}$ and N_{ℓ_y} terms in the summand is called the “layer” of W . For any W , the norms of the commutators of terms in T , U , and V with W are bounded by the ∞ -norms of t , u , and g , and the 1-norm of w .

Lemma 14 ([31, Proposition F.4]). *Let an electronic-structure Hamiltonian be given as in Eq. (B4). The following statements hold for a general second-quantized operator W with q layers:*

1. $\tilde{W} = [T, W]$ is an operator with q layers and $\|\vec{w}\|_1 \leq 2qn\|\vec{t}\|_\infty\|\vec{w}\|_1$;
2. $\tilde{W} = [U, W]$ is an operator with q layers and $\|\vec{w}\|_1 \leq 2q\|\vec{u}\|_\infty\|\vec{w}\|_1$; and
3. $\tilde{W} = [V, W]$ is an operator with $q + 1$ layers and $\|\vec{w}\|_1 \leq 4qn\|\vec{g}\|_\infty\|\vec{w}\|_1$.

A Corollary of this lemma is an upper bound for the sum of right-nested commutator norms.

Corollary 2. *The sum of the norms of the right-nested commutators of H_v satisfies*

$$\sum_{v_1, \dots, v_q=1}^M \|[H_{v_1}, \dots, H_{v_q}]\| = O(q!n^q),$$

for any $q \geq 2$. In addition, each $[H_{v_1}, \dots, H_{v_q}]$ has at most q layers.

Proof. We prove the statement by induction on q .

Consider the base case $q = 2$. The LHS is $\sum_{v_1, v_2=1}^M \|[H_{v_1}, H_{v_2}]\|$, where each H_{v_1} and H_{v_2} is a term belonging to T , U , or V . Note that terms from U and V commute with each other, so we only need to consider commutators between T and $\{U, V\}$. This yields

$$\sum_{v_1, v_2=1}^M \|[H_{v_1}, H_{v_2}]\|$$

$$\begin{aligned}&= O(n\|\vec{t}\|_\infty\|\vec{u}\|_1 + n\|\vec{t}\|_\infty\|\vec{g}\|_1 + n\|\vec{t}\|_\infty\|\vec{t}\|_1) \\ &= O(n^2).\end{aligned}$$

The first equality follows from Lemma 14 and the second follows from the coefficient norms in Lemma 13. In addition, the layer of $[H_{v_1}, H_{v_2}]$ is at most 2. Therefore, the base case holds.

Assume the statement holds for $q - 1$. Consider the sum for q :

$$\begin{aligned}&\sum_{v_1, \dots, v_q=1}^M \|[H_{v_1}, \dots, H_{v_q}]\| \\ &= \sum_{v_1=1}^M \sum_{v_2, \dots, v_q=1}^M \|[H_{v_1}, [H_{v_2}, \dots, H_{v_q}]]\|.\end{aligned}$$

By the induction hypothesis, $[H_{v_2}, \dots, H_{v_q}]$ has at most $q - 1$ layers. According to Lemma 14, commuting with terms belonging to T , U , or V increases the norm by a factor proportional to $(q - 1)n\|\vec{t}\|_\infty = O(q)$, $(q - 1)\|\vec{u}\|_\infty = O(qn)$, or $(q - 1)n\|\vec{g}\|_\infty = O(qn)$, respectively. Summing over all possible H_{v_1} , we have

$$\begin{aligned}&\sum_{v_1=1}^M \sum_{v_2, \dots, v_q=1}^M \|[H_{v_1}, [H_{v_2}, \dots, H_{v_q}]]\| \\ &= O\left(qn \sum_{v_2, \dots, v_q=1}^M \|[H_{v_2}, \dots, H_{v_q}]\|\right) \\ &= O(q!n^q),\end{aligned}$$

where the last equality follows from the induction hypothesis. By Lemma 14, commuting with an H_v increases the layer by at most one, and hence $[H_{v_1}, \dots, H_{v_q}]$ has at most q layers. $\square$

Let $\mathcal{K}_v := -\text{iad}_{H_v}$ and

$$\mathcal{K} := \sum_{v=1}^M \mathcal{K}_v = -\text{iad}_H. \quad (\text{B5})$$

We now apply Theorem 1 to bound the truncation error of the BCH formula of $\prod_{v=1}^M \mathcal{K}_v$. The following lemma gives the required doubly right-nested commutator bound for the second-quantized electronic-structure Hamiltonian.

Lemma 15. *Let $\mathcal{K} = \sum_{v=1}^M \mathcal{K}_v$ be the Liouvillian in Eq. (B5). For a given sequence of positive integers $q_1, q_2, \dots, q_d$, let $P_r = \sum_{j=r}^d q_j$ for $r = 1, 2, \dots, d$ and $P_{d+1} = 1$. Then, the doubly right-nested commutator bound $\bar{\alpha}_{\text{comm}}^{(q_1, \dots, q_d)}$ for $\mathcal{K}$ (see Eq. (8)) satisfies*

$$\bar{\alpha}_{\text{comm}}^{(q_1, \dots, q_d)} = O\left(\prod_{r=1}^d P_{r+1} q_r! n^{q_r}\right). \quad (\text{B6})$$

Proof. First, we reduce the problem to bounding commutators of Hamiltonian terms. By applying the Jacobi identity $[\text{ad}_X, \text{ad}_Y] = \text{ad}_{[X, Y]}$ recursively and using $\|\text{ad}_X\| \leq 2\|X\|$, the doubly right-nested commutator bound of $\mathcal{K}_v$ satisfies

$$\begin{aligned} & \bar{\alpha}_{\text{comm}}^{(q_1, \dots, q_d)} \\ &= \sum_{v_{1,1}, \dots, v_{d,q_d}=1}^M \left\| \text{ad}_{[H_{v_{1,1}}, \dots, H_{v_{1,q_1}}], \dots, [H_{v_{d,1}}, \dots, H_{v_{d,q_d}}]} \right\| \\ &\leq 2 \sum_{v_{1,1}, \dots, v_{d,q_d}=1}^M \left\| [H_{v_{1,1}}, \dots, H_{v_{1,q_1}}], \dots, [H_{v_{d,1}}, \dots, H_{v_{d,q_d}}] \right\| \end{aligned}$$

Therefore, it suffices to bound the sum of doubly right-nested commutators of H_v .

To simplify the notation, for any layer $r \in [d]$, let $\vec{v}_r = (v_{r,1}, \dots, v_{r,q_r})$ denote the index vector, and define the nested commutator for that layer as

$$H_{\vec{v}_r} := [H_{v_{r,1}}, \dots, H_{v_{r,q_r}}].$$

We define the partial nested commutator starting from layer r as

$$W_{\vec{v}_r, \dots, \vec{v}_d} := [H_{\vec{v}_r}, \dots, H_{\vec{v}_{d-1}}, H_{\vec{v}_d}].$$

We prove the following bound by backward induction on r from d to 1:

$$\sum_{\vec{v}_r, \dots, \vec{v}_d} \|W_{\vec{v}_r, \dots, \vec{v}_d}\| = \mathcal{O}\left(\prod_{j=r}^d P_{j+1} q_j! n^{q_j}\right). \quad (\text{B7})$$

In addition, we show that $W_{\vec{v}_r, \dots, \vec{v}_d}$ has at most P_r layers.

Consider the base case $r = d$. By [Corollary 2](#), the LHS of Eq. (B7) is

$$\sum_{\vec{v}_d} \|H_{\vec{v}_d}\| = \mathcal{O}(q_d! n^{q_d}) = \mathcal{O}(P_{d+1} q_d! n^{q_d}),$$

and $H_{\vec{v}_d}$ has q_d layers, so the base case holds.

Assume Eq. (B7) holds for r . We consider the sum for $r-1$:

$$\sum_{\vec{v}_{r-1}, \dots, \vec{v}_d} \|W_{\vec{v}_{r-1}, \dots, \vec{v}_d}\| = \sum_{\vec{v}_r, \dots, \vec{v}_d} \sum_{\vec{v}_{r-1}} \left\| [H_{\vec{v}_{r-1}}, W_{\vec{v}_r, \dots, \vec{v}_d}] \right\|.$$

If $q_{r-1} \geq 2$, we have

$$\begin{aligned} & \sum_{\vec{v}_r, \dots, \vec{v}_d} \sum_{\vec{v}_{r-1}} \left\| [H_{\vec{v}_{r-1}}, W_{\vec{v}_r, \dots, \vec{v}_d}] \right\| \\ &\leq 2 \left(\sum_{\vec{v}_{r-1}} \|H_{\vec{v}_{r-1}}\| \right) \left(\sum_{\vec{v}_r, \dots, \vec{v}_d} \|W_{\vec{v}_r, \dots, \vec{v}_d}\| \right) \\ &= \mathcal{O}\left(q_{r-1}! n^{q_{r-1}} \prod_{j=r}^d P_{j+1} q_j! n^{q_j}\right) \end{aligned}$$

$$= \mathcal{O}\left(\prod_{j=r-1}^d P_{j+1} q_j! n^{q_j}\right),$$

where the second line follows from [Corollary 2](#) and the induction hypothesis. The layer count of $W_{\vec{v}_{r-1}, \dots, \vec{v}_d}$ is the sum of the layer counts of $H_{\vec{v}_{r-1}}$ and $W_{\vec{v}_r, \dots, \vec{v}_d}$, which is $q_{r-1} + P_r = P_{r-1}$.

Then we consider the case of $q_{r-1} = 1$. We have

$$\sum_{\vec{v}_{r-1}, \dots, \vec{v}_d} \|W_{\vec{v}_{r-1}, \dots, \vec{v}_d}\| = \sum_v \sum_{\vec{v}_r, \dots, \vec{v}_d} \left\| [H_v, W_{\vec{v}_r, \dots, \vec{v}_d}] \right\|.$$

By the induction hypothesis, $W_{\vec{v}_r, \dots, \vec{v}_d}$ has at most P_r layers. Using the same argument in the proof of [Corollary 2](#), commuting with an H_v increases the norm by a factor of $\mathcal{O}(P_r n)$. Therefore, we have

$$\begin{aligned} & \sum_v \sum_{\vec{v}_r, \dots, \vec{v}_d} \left\| [H_v, W_{\vec{v}_r, \dots, \vec{v}_d}] \right\| \\ &= \mathcal{O}\left(P_r n \sum_{\vec{v}_r, \dots, \vec{v}_d} \|W_{\vec{v}_r, \dots, \vec{v}_d}\|\right) \\ &= \mathcal{O}\left(P_r n \prod_{j=r}^d P_{j+1} q_j! n^{q_j}\right) \\ &= \mathcal{O}\left(\prod_{j=r-1}^d P_{j+1} q_j! n^{q_j}\right), \end{aligned}$$

where the second line follows from the induction hypothesis. By [Lemma 14](#), $W_{\vec{v}_{r-1}, \dots, \vec{v}_d}$ has at most $1 + P_r = P_{r-1}$ layers. This completes the induction. Setting $r = 1$ yields the lemma statement. $\square$

Appendix C: Experimental Details

The hardware data are processed through the following pipeline. We first collect raw computational-basis bit-strings and apply observable-level readout mitigation to estimate the magnetization. Cliffordized training circuits are then used to learn a layerwise sparse Pauli-Lindblad model for the compiled device circuit. The learned model determines random Pauli insertions that realize several amplified noise levels. For each fixed Trotter step number, these data are extrapolated to zero physical noise. Finally, the zero-noise estimates at different Trotter step numbers are extrapolated in $x = 1/N_{\text{Trotter}}$ to reduce the deterministic product-formula error.

1. Readout Error Calibration

The hardware experiment uses a connected 9-qubit subgraph of a 13-qubit superconducting processor. Before learning the effective gate-noise channel, we first calibrate and mitigate measurement errors. Since the target observable is the average longitudinal magnetization,

$O = n^{-1} \sum_i Z_i$, readout correction can be applied directly at the observable-estimation level.

Figure S1 shows both the full $2^9 \times 2^9$ confusion matrix on the selected subgraph and the corresponding single-qubit calibration matrices. The global matrix is strongly diagonally dominant, indicating that the readout channel is close to an independent bit-flip model on the relevant device subgraph. The local calibration matrices therefore capture the dominant readout contribution for the magnetization observable, while avoiding the sampling cost of inverting the full global matrix in every data-processing step.

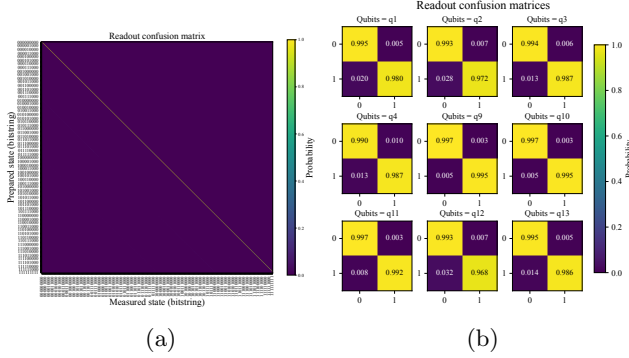


FIG. S1. Readout calibration data for the selected 9-qubit device subgraph. (a) Full $2^9 \times 2^9$ global confusion matrix, whose entries give the conditional probabilities of measured bit-strings given prepared computational-basis states. (b) Single-qubit local calibration matrices used for observable-level readout mitigation of the magnetization. The diagonal dominance of the calibration data supports the use of local observable-level correction for the measured average magnetization.

2. Noise Channel Learning

The learned-channel quantum error mitigation workflow assumes an effective sparse Pauli–Lindblad model for each relevant circuit layer [20],

$$\mathcal{L}(\rho) = \sum_{k \in \mathcal{K}} \lambda_k (P_k \rho P_k - \rho),$$

where $\mathcal{K}$ contains the Pauli components retained by the sparse model. The corresponding noise channel is diagonal in the Pauli basis, so scaling the rates λ_k provides a controlled way to generate larger effective noise strengths for ZNE.

To learn these rates on hardware, we construct Clifford training circuits that are structurally matched to the target Trotter circuits. The training circuits preserve the same qubit mapping, connectivity-respecting gate schedule, layer grouping, and native-gate compilation, but replace the non-Clifford rotation angles by Clifford angles in $\{0, \pi/2, \pi, 3\pi/2\}$. Their noiseless expectation values can be computed classically, while their noisy expectation values are measured on the chip and corrected for

readout error. Comparing these two sets of data gives a sparse inverse problem whose solution yields the effective Pauli–Lindblad rates for each Clifford layer.

Figure S2 shows a representative learned hardware noise channel for a Trotterized circuit with step number $N = 3$. The layer-resolved structure of the inferred Pauli components provides a direct diagnostic that the learning stage is extracting a structured effective noise description for the compiled circuits, rather than fitting only a single aggregate depolarizing rate.

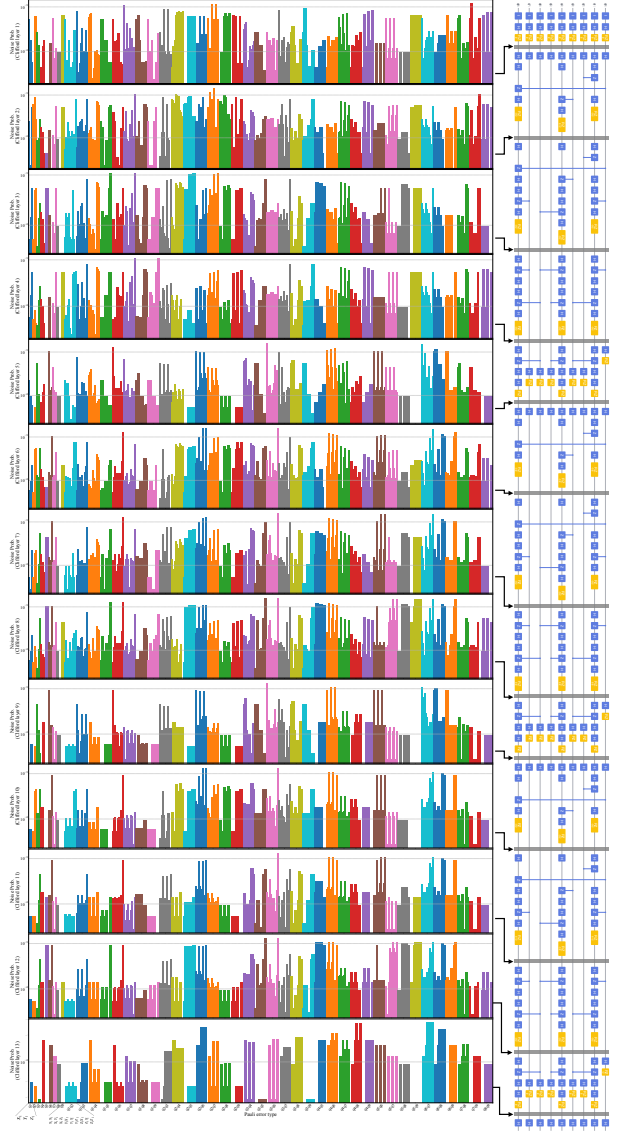


FIG. S2. Learned effective hardware noise model for a Trotterized Ising circuit with Trotter step number $N = 3$. The plot shows the inferred sparse Pauli–Lindblad components associated with the Clifford layers in the compiled hardware circuit; each bar corresponds to one retained Pauli error channel. The layer-resolved structure is the model used to generate controlled noise amplification in the subsequent ZNE step.

3. Zero-Noise Extrapolation Workflow

Figure S3 illustrates the overall experimental learning-and-mitigation workflow. Target Trotter circuits are first mapped to the hardware topology and compiled into native gates. Adjacent Clifford operations are then grouped into effective Clifford layers. The Cliffordized training circuits use the same compiled layout as the target circuits, but restrict the rotation angles so that the resulting circuits remain classically simulable. Hardware measurements of these training circuits are combined with the noiseless Clifford references to learn an effective layerwise sparse Pauli–Lindblad noise model. This learned model is subsequently used to implement controlled noise amplification by random Pauli insertions.

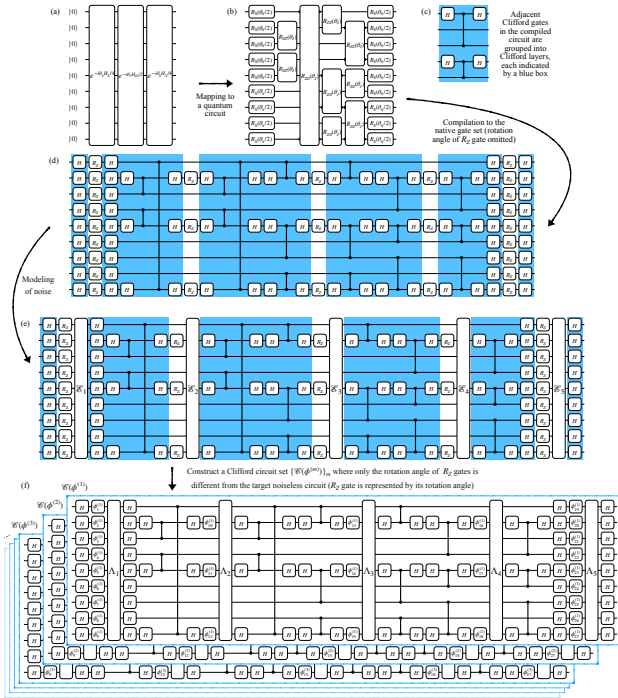


FIG. S3. Experimental learning-and-mitigation workflow. Target Trotter circuits are mapped to the hardware topology and compiled into native gates. Cliffordized training circuits preserve the same layer structure while replacing rotation angles by Clifford angles. Hardware measurements of these training circuits, together with noiseless Clifford references, determine an effective layerwise sparse Pauli–Lindblad model used for noise amplification and ZNE. The workflow keeps the training circuits structurally matched to the target Trotter circuits so that the learned channel is tied to the compiled hardware implementation.

After learning the effective noise channel, we use ran-

dom Pauli insertions to implement gain factors for the noise strength. For each fixed Trotter step number, the amplified-noise expectation values are fitted as functions of the gain factor and extrapolated to the zero-noise limit. In the hardware data we compare low-order polynomial fits with exponential fitting families; the latter are natural for Pauli–Lindblad channels because the channel eigenvalues depend exponentially on the learned rates.

Figure S4 shows hardware ZNE data for two representative evolution settings. In both cases, the measured expectation values vary smoothly with the gain factor, and the zero-noise estimates are consistent with the corresponding noiseless simulation references within the accuracy of the fit.

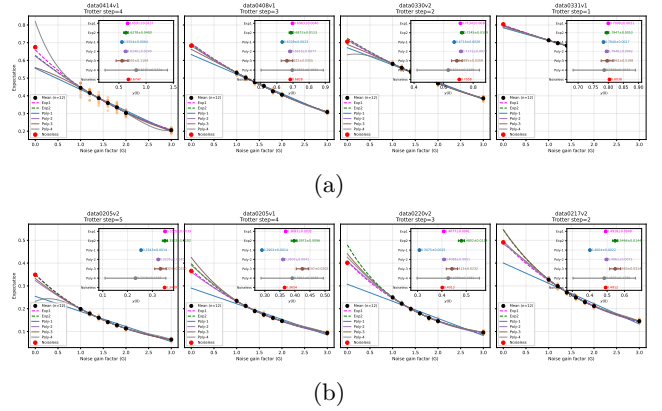


FIG. S4. Hardware zero-noise extrapolation using the learned Pauli–Lindblad channel. (a) Evolution setting $r_x T = r_z T = -1$. (b) Evolution setting $r_x T = r_z T = -2$. The extrapolated value at zero noise is obtained by fitting the measured dependence on the gain factor. These data provide the finite-Trotter, zero-noise estimates that are subsequently used for Trotter-step extrapolation.

The hardware ZNE stage removes the physical-noise bias at fixed Trotter step number. It does not remove the deterministic error from the finite-step product formula. For this reason, the ZNE-corrected estimates are subsequently extrapolated in $x = 1/N_{\text{Trotter}}$. Because the hardware circuit uses a second-order symmetric Trotter formula, the leading product-formula corrections have an even-power structure in x , motivating the even-polynomial fits used in the main text. The combined procedure therefore treats stochastic hardware noise and deterministic Trotter error in two compatible extrapolation stages.

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